



Pós-Graduação em Ciência da Computação

**AN APPLICATION OF DATA MINING TO THE PROGRAM
RECIFE CITY COUNCIL SCHOOL GRANT**

put

ROBERTO TABOSA FLORENCIO FILHO

Master's Dissertation



Federal University of Pernambuco

posgraduacao@cin.ufpe.br

www.cin.ufpe.br/~posgraduacao

RECIFE, APRIL 2009



Federal University of Pernambuco

COMPUTER CENTER

POSTGRADUATE DEGREE IN COMPUTER SCIENCE

ROBERTO TABOSA FLORENCIO FILHO

**AN APPLICATION OF DATA MINING TO THE PROGRAM
RECIFE CITY COUNCIL SCHOOL GRANT**

Supervisor: Prof. Dr. Paulo Jorge Leitão Adeodato

Dissertation presented to the Postgraduate Program in
Computer Science at the Computer Science Center of the
Federal University of Pernambuco as a partial requirement for
obtaining the degree of Master in Computer Science.

RECIFE, APRIL 2009

Roberto Tabosa , Florencio Filho

An application of data mining to the school grant program of the city of Recife / Roberto Tabosa Florencio Filho. - Recife: The Author, 2009. **ix, 100 pages** :
ill., fig., tab., tables, graph.

Dissertation (master's degree) Federal University of Pernambuco. Cln. Computer Science, 2009.

Includes bibliography and appendix.

1. Data mining. 2. Artificial intelligence. 3. Artificial neural networks. I. Title.

006.312

CDD (22nd ed.)

MEI2010 050

ACKNOWLEDGMENTS

I thank all those who in some way participated, encouraged me and rooted for the success of this work. In a special way, my sincere thanks go thanks:

To God, for having granted me strength , health and a family wonderful to be able to reach the conclusion of this work.

To my father, in memory, an example of character, perseverance and dignity, who has always been my reference.

To my wife for her understanding and support in the most difficult and uncertain times. To my daughter for existing and being a natural force for me to overcome difficulties encountered.

To my mother and brothers for their support and words of encouragement.

To my advisor, Professor Paulo Adeodato, for the opportunities given to me given, for the criticisms and mainly for the firm stance in demanding activities.

To my friend and boss, Moacir Antônio Marafon, for the understanding required during this journey.

To my friend José Antônio Manso, for having managed to get the City Hall to from Recife and EMPREL the database so that this work could be carried out.

To colleagues David Fernandes França and Fabiano N. de Carvalho, for the important exchange of knowledge during the discipline of decision support systems and data mining.

To Tereza Sato and Rafaela Ávila, for their attention and availability whenever I needed clarification about the Bolsa Escola Program of the City Hall of Reef.

SUMMARY

AN APPLICATION OF DATA MINING TO THE PROGRAM RECIFE CITY COUNCIL SCHOOL GRANT

The Data Mining task involves a set of statistical techniques and artificial intelligence with the aim of discovering information not found by tools commonly used for extracting and storing data in large databases. Data Mining can be applied to any area of knowledge (Exact Sciences, Humanities, Social Sciences, Biological Sciences, Health Sciences, Agriculture and others) providing gains in information and knowledge, currently unknown, in any of them.

This work presents an application of data mining to the Bolsa program School of the City of Recife (PCR), particularly in the investigation of situation of the beneficiary families, with the aim of offering the administration municipal a decision support tool capable of improving the decision-making process granting of benefits. A mass of socio-economic data was analyzed initially of around 60 thousand families registered in the program. A MultiLayer Perceptron (MLP) artificial neural network to classify families benefited based on their socio-economic characteristics .

The evaluation of performance and results obtained, in addition to the response from expert in the application domain, demonstrate the viability of this application in process of granting the benefit to the Bolsa Escola Program of the City Hall from Recife.

Keywords: Knowledge Discovery in Databases, Cross-Industry Standard Process for Data Mining (CRISP-DM), Data Mining, Neural Networks, Municipal School Grant Program (PBEM).

ABSTRACT

A DATA MINING APPLICATION TO THE RECIFE CITY COUNCIL BOLSA ESCOLA PROGRAM.

The Data Mining task involves a set of statistical techniques and artificial intelligence aiming to discover information which were not found by tools usually used for data extraction and storage in large databases. The Data Application Mining can be done in any area of knowledge, such as Exact, Human, Social, Biological, Health and Agrarian Sciences, and others, providing information and knowledge gaining, now unknown, in any one of them.

This study presents an application of data mining to the Recife City Council Bolsa Escola Program, particularly in the inquiry of the benefited families situation, with the objective of offering the municipal administration a decision support tool able to improve the concession process of benefits. A social-economic data mass was initially analyzed in about 60 thousand families registered in the program. An artificial neural net MultiLayer Perceptron (MLP) was used to classify the benefited families based on their social-economic characteristics.

Performance evaluation and the obtained results, besides the specialist answer in the application mastery, showed its viability in the benefit concession process of the Recife City Council Scholarship Program.

Keywords: Knowledge Discovery in databases, Cross-Industry, Standard Process for Data Mining (CRISP-DM), data mining, neural networks, Council Bolsa Escola Program.

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MAIN ABBREVIATIONS

CRISP-DM	<i>Cross-Industry Standard Process for Data Mining</i>
CSR	Representative Social Cost
DASE	Directorate of Social Support for Education
ECA	Statute of Children and Adolescents
IBGE	Brazilian Institute of Geography and Statistics
HDI	Human Development Index
KDD	<i>knowledge discovery in databases</i>
KS	<i>Kolmogorov-Smirnov</i>
MLP	<i>Multilayer Perceptron</i>
MSE	<i>Mean Squared Error</i>
UN	United Nations
PBEM	Municipal School Grant Program
PCR	City Hall of Recife
PNAD	National Household Sample Survey
UNDP	United Nations Development Programme
RNA	Artificial Neural Networks
ROC	<i>Receiver Operating Characteristics</i>
SM	Minimum wage

1 Introduction

Initially proposed by the federal government in the 1990s, the programs minimum income and school grants are one of the most significant phenomena in the scenario of social policies in Brazil.

Present in several units of the federation, these programs have in common the family focus and the link with early childhood education, as well as the perception that granting a monetary benefit yields better results than policies traditional assistance [AR99].

This is mainly because many families are unable to maintain children at school due to their low quality of life and extreme poverty; and, with insufficient income, they rely on children's work for their livelihood, characterizing child labor as one of the main impediments to school attendance [CS03].

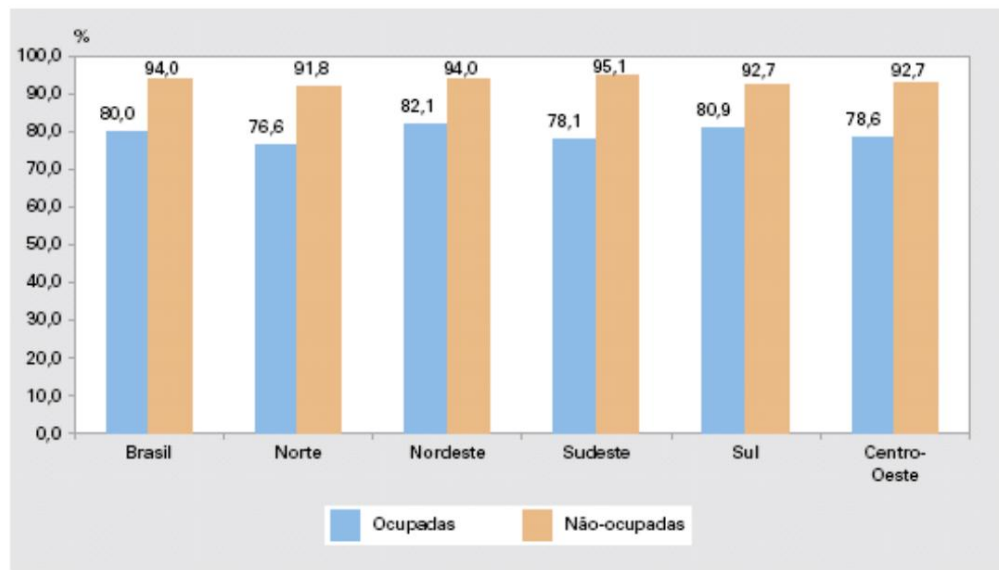
Data from the National Household Sample Survey (PNAD)¹ published in 2007 [PNAD07] reinforce this statement, showing that in all Regions of the Brazil, school enrollment rates are higher in groups of people who do not work (Graph 1).

Statistical studies [MS99] show that in general, children who have parents illiterate, without guaranteed income, insufficient food, lack of energy, water plumbing and sewage, among other survival problems, have their learning committed, compared to others that do not have the same degree of poverty.

1.1 Motivation and Justification

Works with a social, economic or educational focus have already been developed about the Bolsa Escola Program in State City Halls

¹ National Household Sample Survey. Conducted annually by the Brazilian Institute of Geography and Statistics (IBGE).



Fonte: IBGE, Diretoria de Pesquisas, Coordenação de Trabalho e Rendimento, Pesquisa Nacional por Amostra de Domicílios 2007.

Graph 1 - Schooling rate of people aged 5 to 17, by employment status in the reference week, in 2007, taken from [PNAD07].

Brazilians; several other works have also been carried out on Mining Data, however, there are no specific studies using Data Mining techniques. Data aimed at assessing the influence of socio-economic characteristics of families benefited by the Municipal School Grant Program (PBEM) of the City Hall of Recife (PCR) in its success.

Despite the financial incentive, some families are unable to maintain their dependents at school and, therefore, cease to be characterized as beneficiaries, harming the program's mission.

Thus, this work was motivated by the academic interest in offering public managers the improvement of the decision-making process for granting the benefit as a way to increase program efficiency and optimize investment intended for the same. For this work the socio-economic characteristics were considered economic conditions of all appropriate applicants and those removed from the Program. Subsequently, the labels Success and Failure were created in a new attribute, to

better characterize each applicant for the Program as will be explained in the section 5.4.2.

1.2 Objectives of the Work

Improve the decision-making process for granting the benefit, offering a decision support tool, using data mining method based on analysis of the available registration data of suitable and rejected applicants , in order to allow program managers to improve the granting of benefits by Bolsa Escola program of the City of Recife.

The result obtained ensures that experts in the field of application, provision of an important instrument as a criterion for choosing new appropriate claimants for the benefit.

1.3 Dissertation Structure

This dissertation is structured in chapters distributed as follows:
follow related:

- Chapter 2 - describes the Brazilian social and educational environment, in addition to detail the functioning of the Bolsa Escola Program of the City Hall of City of Recife.
- Chapter 3 presents concepts about knowledge discovery in databases.
- Chapter 4 presents concepts about Artificial Neural Networks.
- Chapter 5 describes, through the CRISP-DM methodology, the processes business understanding, data understanding and preparation applied to the Bolsa Escola Program.

- Chapter 6 describes, through the CRISP-DM methodology, the processes of modeling and evaluation applied to the Bolsa Program database School.
- Chapter 7 - concludes the work by presenting the conclusions, limitations existing and suggestions for future work.

2 Economic Situation, Education and the Scholarship Program

This chapter describes the socio-economic situation surrounding education and reports on the operation and existing structure to support the Bolsa Escola Program City Hall of Recife.

2.1 Economic Situation and Education

Social inclusion programs through income transfer are some of the ways found by governments to minimize problems linked to:

- Social exclusion;
- Social need and
- Extreme poverty.

2.1.1 Education - Basic Right

Education in Brazil is a fundamental right guaranteed to every citizen Brazilian by the 1988 Constitution, in its article 6[Const98].

Paragraphs I, II and III of Article 208 (The State's duty to education shall be made effective through the guarantee of:) of the Brazilian Federal Constitution, legislate on the how the Public Authorities should proceed to offer basic education quality to all citizens [Const98]. The Statute of Children and Adolescents (ECA) Law 8,069/90, also provides in article 54 of item VII, paragraph 3, obligation of the Public Authorities to provide due access to primary education through supplementary programs of teaching materials, transportation, food and health care [ECA09]. ”

This right, in recent decades, has occupied a prominent place among the human rights, being considered an essential and indispensable right for the exercise of citizenship. This prominent place within the scope of public power is given essentially for three reasons:

- None of the other rights, whether civil, political, economic or social, can be properly exercised without a minimum of education;
- Education reduces social inequalities, which is, according to the Bank World [TWB44], one of the impact factors for poverty reduction;
- Education is one of the three dimensions that make up the formula for calculating the Human Development Index (HDI) for classifying United Nations (UN) as demonstrated in section 2.1.5 [UNDP06].

Despite all the commitments made by the rulers through international instruments, basic education in Brazil, in general, needs incentives for its effective improvement and social programs through income transfer are a type of action that has been adopted.

2.1.2 Social Exclusion

Social exclusion concerns disadvantaged people or groups. It is the phase extreme of the marginalization process, this being the maximum attainable point where the individual, gradually, moves away from society as a result of ruptures consecutive with the same [Cos98].

A single characteristic does not always determine the situation of social exclusion, for example, the situation in which a low-income family may be considered poor, but be integrated into a community group. A situation (poverty) can lead to another (social exclusion), but not necessarily leading to this.

2.1.3 Social Need

Karl Marx² described social need as something that can only be affirmed through pressure exerted on individuals, with the aim that their decisions have a certain orientation. In this context, the social need is strongly linked to the need for social inclusion and the guarantee of basic social rights guaranteed by law³ : education, health, work, leisure, among others.

2.1.4 Extreme Poverty

In these conditions there are individuals whose basic needs for human survival are not met.

In an improved scenario, there is moderate poverty, when basic needs are met, but individuals remain deprived of basic social needs such as entertainment, health and education.

The methodology used to define poverty in Brazil has the salary minimum as a reference. The following classifications are considered [SEPLAG08]:

- Extremely poor families, those with per capita income of up to $\frac{1}{4}$ minimum wage;
- Poor families, those with per capita income of up to $\frac{1}{2}$ of the salary minimum;

2.1.5 Human Development Index

The Human Development Index (HDI) is a comparative measure that contemplates three dimensions: wealth, education and longevity. It was created in 1990 by

² Karl Heinrich Marx Born in Trier, Germany, on May 5 , 1818 , he was a German intellectual and revolutionary , founder of the modern communist doctrine. Marx's thinking still influences several areas of knowledge today .

³ Basic Rights present in Article 6 of the Federal Constitution of 1988.

Pakistani economist Mahbub ul Haq. He later had the collaboration of Indian economist Amartya Sen, winner of the 1998 Nobel Prize in Economics and has been used since 1993 by the United Nations Development Programme development of its annual report. UN member countries, annually receive a classification according to the HDI [HDI08].

The last annual HDI report was published in 2008 using as a source of data for the survey, updated population data from 2006. According to the research, Brazil is in seventieth place among 179 countries, with an HDI of 0.807 on a scale from 0 to 1. According to the report, the country has an estimated life expectancy around seventy-two years of age and adult literacy rate (above fifteen years of age) of 89.6% [HDR08].

The most recent National Household Sample Survey, with data from year 2007, shows that, at a Brazilian level, in the group of children and adolescents aged 6 to 14 years, the percentage of school attendance was 97%, with 79.2% corresponding to public school education. The research also reveals that the highest schooling rates are in the age groups of 7 to 9 years and 10 to 14 years with 98.1% and 97.4%, respectively [PNAD07] which demonstrates the large concentration of children who, somehow, they are committed to their studies and on the other hand, a window of opportunity for the public authorities to generate better conditions for these children and Young people do not abandon their studies.

In this way, it is clear to see the granting of the benefit of the Bolsa Program School of the City of Recife as a Social Inclusion Program, a since, in addition to the transfer of income providing better economic conditions, meets social needs by carrying out dozens of socio-educational works with beneficiary families⁴.

⁴ Basic Social Systems: They are grouped into five domains: -social, economic, institutional, territorial and symbolic references.

2.2 The School Grant Program

The Bolsa Escola Program of the City of Recife is a program of income transfer and social inclusion, created by Law 16,302 and regulated by Decree 17,665 in 1997, conducted by the Directorate of Social Support for Education (DASE), linked to the Secretariat of Education, Sports and Leisure, having as its main aim to serve needy families, this service being conditional on enrollment and children's permanence in a municipal school in Recife.

2.2.1 Objective of the Bolsa Escola Program

The main objective of the Program is to reduce dropout rates in schools municipal. The families benefiting receive a financial incentive to motivate them their children's school attendance. The benefit amount is half a salary current minimum (currently two hundred and thirty-two reais and fifty cents) for who has a child in school or a current minimum wage (currently four hundred and sixty-five reais) for those who have two or more children at school. This amount is paid only by the City of Recife. In this context, benefit value is characterized as financial compensation to guarantee the children's permanence in school. Currently , an average of 1.9 million euros are invested reais per month for payment to beneficiary families [PCR08].

2.2.2 Operation of the Bolsa Escola Program

The information in this section was obtained through interviews and reports delivered by the Directorate of Social Support for Education (DASE) of the PCR.

All criteria for including a family in the program must be satisfied. They are:

- The applicant must have completed the registration;
- Families with children aged six to fourteen years and eleven months;
- Families with per capita income of up to 1/3 of the current minimum wage;

- Children enrolled who study in schools of the Municipal Network of Reef;
- Families that have lived in Recife for at least five consecutive years;

In 2001, 60,451 families were registered. The data for these families are used to date to make new inclusions to the benefit, with no new registrations, since there are several families to be considered and that wait for the benefit.

The process of granting the benefit follows a series of steps that are described below. The data provided at the time of registration of the interested party (applicant) are stored in a database in which the SEBE system⁵, from this data, generates a score⁶ and applicants considered most in need receive a higher score and those less in need receive a higher score lowest. Some of the applicant's characteristics and their respective scores are listed in Appendix A. If financial availability exists, the largest scored are listed and visited to confirm the information. If it continues with the profile, a final evaluation of the candidate family is made by a committee, to finally proceed with the effective implementation of the classified applicants who, with this, become receive the benefit earned monthly. The inclusion of the applicant will pass his/her situation from scored to appropriate . ”

To remain in the Program, in addition to continuing to meet the criteria for inclusion the applicant must be living with the child (dependent), justify the absences from school, monitor the child's development at school, use adequately benefit and participate in the Program's activities.

⁵ System developed in 2001 for the Municipal School Grant Program (PBEM), through a partnership between the Department of Education, Sports and Leisure (SEEL) and the Municipal IT Company (EMPREL).

⁶ The applicant's SEBE score is given by the sum of the scores for each socioeconomic characteristic recorded and confirmed after the interviewer's visit. Some characteristics are shown in Appendix A.

If any of the criteria are no longer met, the family (applicant) will be removed from the Program and its status in the database changed from appropriate to away. The exception is in the case of applicants with more than one dependent and some of these dependents cease to meet the conditions for remaining in the Program due to age (upon reaching 14 years and 12 months); for this situation, only the dependent(s) are removed from the Program, but the family (applicant) continues receiving the benefit. The structure and operation of DASE are described in

Illustration 1:

Organization chart:

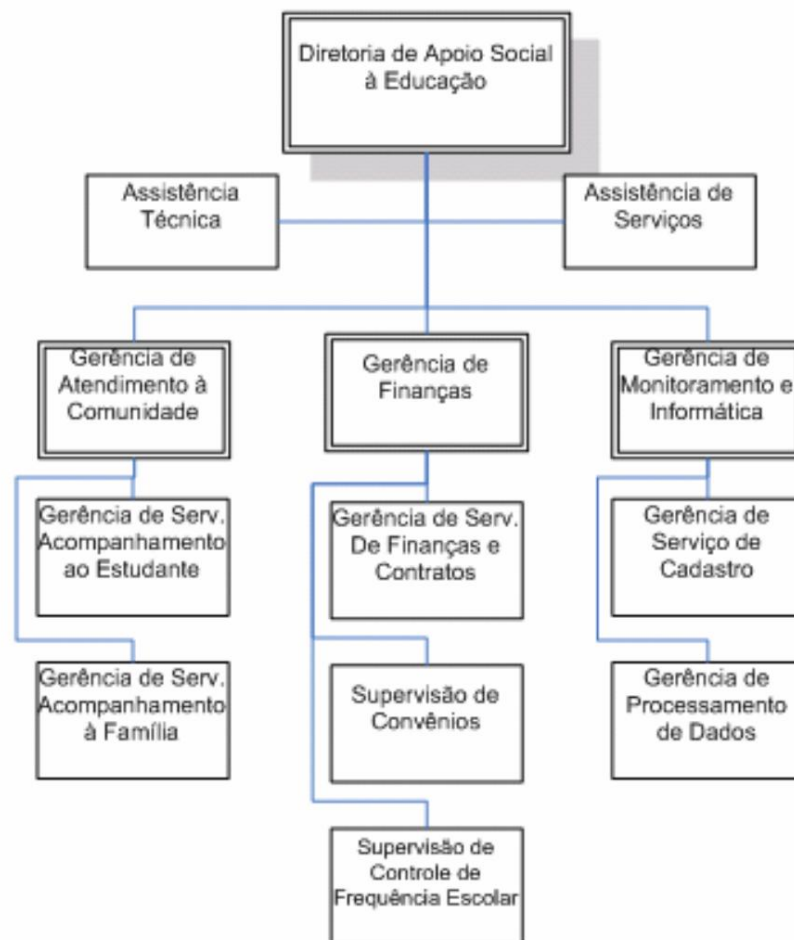


Illustration 1 - Organizational structure chart for controlling Bolsa Escola

Responsibilities:

Community Service Management responsible for inclusion and monitoring of beneficiary families, assessing the socio-economic situation and the compliance with the criteria established for remaining in the Program, carrying out visits home visits and assistance to applicants and beneficiaries at the Directorate itself.

Student Support Service Management — makes visits home to applicants and beneficiaries, for inclusion, monitoring of dependents, investigation of complaints and response to requests from the Councils Guardianships, Municipal Schools, Juvenile Court, NGOs, among others.

Family Support Service Management provides assistance to public, providing information about the Program's criteria, about the importance of good use of the benefit granted, updating registration information and positioning the applicants about the current status of their registration.

Finance Management responsible for the payroll of beneficiaries of the Program and all the steps that precede payment, in addition to monthly control of the school attendance.

Finance and Contracts Services Management makes changes to the payroll payment due to inclusion of new beneficiaries and terminations; carries out also changes in the amount granted to the beneficiary and requests blocks and unlocking of Caixa Econômica Federal monthly.

Agreement Supervision validates changes made to the payroll payment monitoring the situation of beneficiaries receiving the benefit of complementary form (Bolsa Família and Bolsa Escola Programs) and when necessary, visiting them.

School Attendance Control Supervision Controls and monitors the municipal school attendance .

Monitoring and IT Management — Responsible for the area of computer science and storage of the Program's registration data through the System School Grant (SEBE) and Payroll (FGET), as well as reporting bimonthly on the Municipal School Grant Program (PBEM).

Registration Service Management — Performs control and archiving of records of all applicants and beneficiaries enrolled in the Program, preparing archiving techniques to facilitate data extraction when requested.

Data Processing Service Management Performs data-entry PBEM registration in the systems, in addition to being responsible for monitoring the maintenance and installation of computers and computer programs.

Technical and Service Assistance Provide direct support to the Support Directorate Social to Education.

The process flow of the Bolsa Escola Program of the City Hall of Recife is shown in Illustration 2.

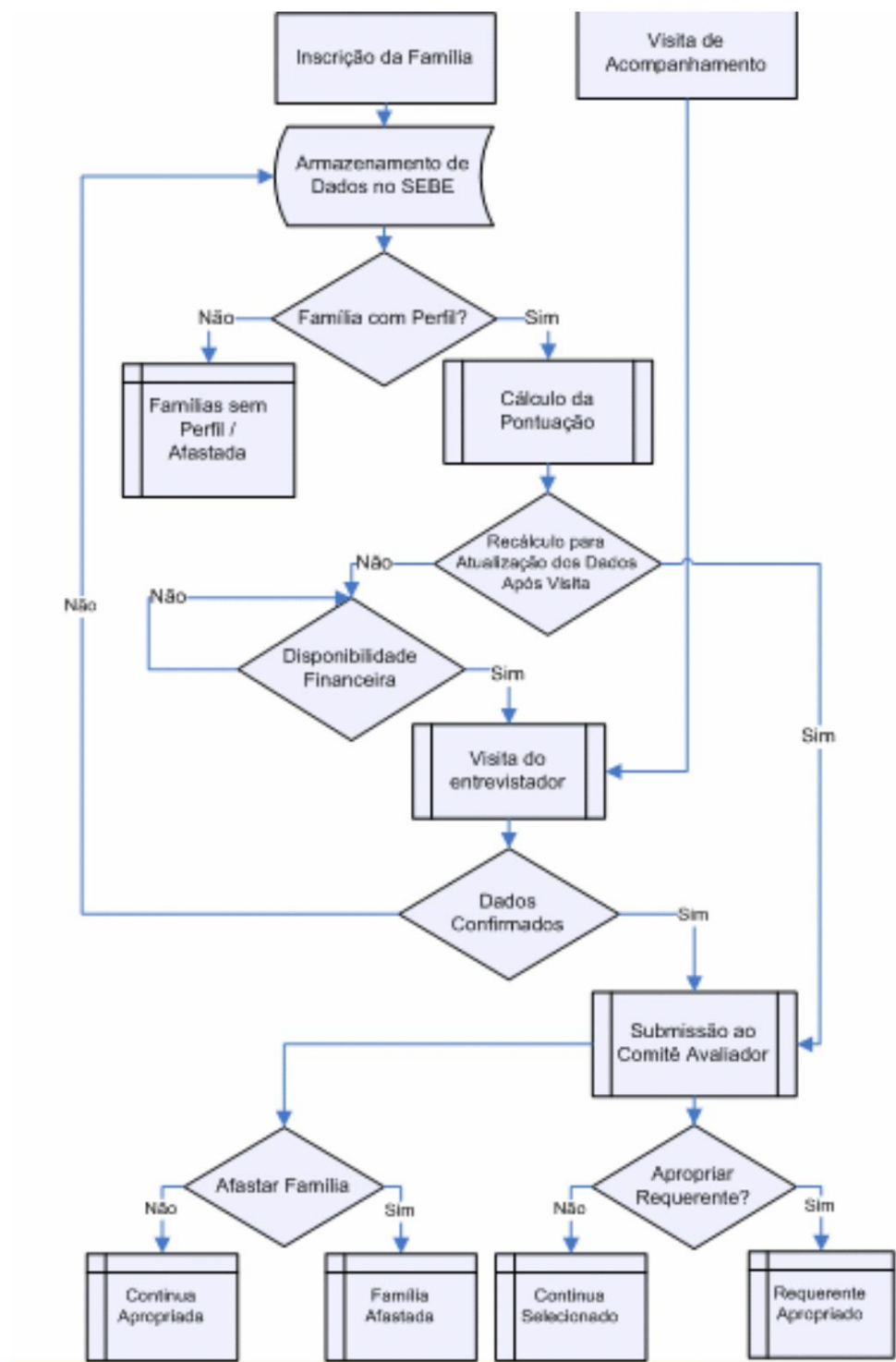


Illustration 2 - Process flow of the PCR Bolsa Escola Program

3 Knowledge Discovery in Databases

Data

This chapter covers the phases of the discovery process.
knowledge through databases.

3.1 Introduction

The process of knowledge discovery in databases was defined by Fayyad, along with other researchers, in 1996 as being a *non-trivial, multi-step process*⁷, *8 interactive*⁸ *9 and iterative*⁹ *for pattern identification* *understandable, valid, novel and potentially useful from large sets of data* [FSS96]. The *Knowledge Discovery in Databases* (KDD) process was defined in five phases: data selection, data pre-processing, transformation, data mining and interpretation of results. The phases of the entire KDD process are shown in Illustration 3 and described in the following sections of this chapter.

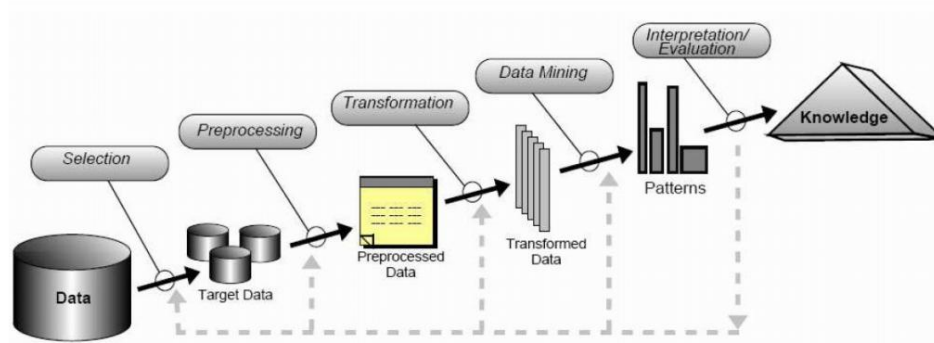


Figure 3 - Phases of the knowledge discovery process in databases [FPS96]

⁷ Non-trivial due to the complexity involved in the problem.

⁸ Interactive because it requires user intervention.

⁹ Iterative because each phase of the entire process can be executed multiple times.

3.2 KDD Phases

3.2.1 Data Selection

At this stage, the data sources are selected, which can be internal sources. or external to the company or even a source of tacit knowledge of experts. These sources can come from different database architectures, the internet, spreadsheets and documents originating from different applications or even documents manuscripts that should be digitized.

3.2.2 Data Preprocessing

At this stage, adjustments are made to the data in order to eliminate the existing inconsistencies. Cases of errors in filling in fields are handled, data outside the expected standard and absence thereof. This phase corrects the database data avoiding the attempt of analysis by the mining algorithm. It demands about eighty percent of the total effort in the KDD process.

The use of a data warehouse facilitates this step because it is inherent to use of data extraction, processing and loading routines which leads to greater level of data consistency, but a KDD project is not rendered unfeasible by the absence of that.

Some examples of data processing would be:

- Null values can be statistically replaced by the mean, when dealing with numerical data; replaced by a label or by the mode, when the data are categorical. Null or missing values can still be filled in manually or have their tuples discarded and finally, you can choose to fill them through data mining methods.
- Standardization of fields; dates or replacing dates with periods (days, months, others.).

- Non-standard values can be corrected by deleting tuples that have values outside the domain or, alternatively, opt for correction of these values manually or via Structured Query Language (SQL).
- Too many samples; consider a higher level of granularity.

The selection of the most relevant attributes for the mining objective also is part of the scope of this stage. This task aims to optimize the time of processing of the data mining algorithm. Two main methods, in addition to the knowledge of the domain expert, are used for attribute selection [AVL99]:

- Multiple iterations: uses the Forward Selection and Backward Selection methods Elimination. The first of these is started with an empty set of attributes and iteratively successively selects one attribute by time, using as a threshold the improvement in the quality measure of result.
- Simple iteration: uses algorithms with statistical foundations distribution probabilities.

The pre-processing phase also suggests data enrichment. This is the creation of new information based on the knowledge of experts in the domain and data, in order to improve the quality of existing information, possibly increasing the performance of the mining algorithm.

3.2.3 Transformation

In this phase, data transformation is carried out, aiming at the need for algorithm to be used in the mining phase. In the case of an Artificial Neural Network, for example, the data needs to be in a numerical representation.

Data transformation can be done in several ways:

- Direct Transformation is the direct replacement of numerical values by categorical. Example of variable sex:

0 -> M

1 -> F

- Discretization is the continuous mapping into intervals; division of a numerical variable, in pre-defined intervals with the help of the expert in the domain. Example of age variable:

0 to 18 years -> Age Range 1

19 to 26 years old -> Age Range 2

27 to 45 years old -> Age Range 3

There is an essential participation of the specialist in the field, defining the criteria for creating classes.

- Binarization is the transformation of categorical values into values represented by 0 or 1. Example of marital status variable:

Married -> 001

Single -> 010

- Normalization is the transformation of the values of each attribute in such a way generating new values in ranges of $[-1;1]$ or $[0;1]$.

3.2.4 Data Mining

It can be considered the most important phase of the KDD process. It is the phase in which that, in fact, the transformation of data into knowledge to be used by the expert in the application domain. This transformation occurs through the appropriate choice of the algorithm that will be able to efficiently extract hidden knowledge in the database for the proposed task. Table 1 presents some possible tasks with KDD and the respective techniques.

KDD Task	Data Mining Techniques
Association	Statistics and Set Theory
Classification	Statistics, Genetic Algorithms, Networks Neural and Decision Tree
Pattern Grouping	Neural Networks and Statistics
Series Forecast Temporals	Neural Networks, Fuzzy Logic and Statistics

Table 1 - KDD tasks and techniques used

Some of these techniques, as they are the most used, are described below:

3.2.5 Interpretation of Results

In this phase the task of interpreting the data mining results is carried out.
for the proposed problem.

Standards are shown as results. These results must be measured and
validated with the expert in the application domain in order to verify the
feasibility of the constructed model.

4 Artificial Neural Networks

This chapter covers the main concepts involved in the use of artificial neural networks.

4.1 Introduction

The first studies on neural networks took place in the 1940s, when some researchers have contributed significantly to the evolution and use of this concept:

- McCulloch and Pitts in 1943 [McP43], when they suggested that networks neural would be computational machines.
- Hebb in 1949, when he defined the first rule of self-learning organized.
- Rosenblatt in 1958, when he proposed the first learning model supervised with the perceptron¹⁰.

According to Haykin [Hay99], a neural network is a distributed processor parallel and consisting of simple processing units, which have natural propensity to store extracted knowledge and make it available for use, being similar to the human brain in two aspects:

- Knowledge is acquired by the network from its environment through a learning process. In this process, a set of examples is presented to the network, which obtains the features for representation of the information provided that will be used to generate answers to the problem.
- Synaptic weights are used to store knowledge acquired.

¹⁰ It is the simplest form of a neural network used for linear pattern classification separable (patterns that are on opposite sides of a hyperplane).

Artificial Neural Networks (ANNs) are distributed parallel systems composed of simple processing units, which are artificial neurons and which in turn calculate certain mathematical functions. These neurons can be arranged in one or more layers and interconnected by connections that are associated with weights; these weights store knowledge perceived by the model and weigh the input received by each of the various neurons of the ANN [BCL07]. It can be grouped into input, intermediate (or hidden) and output layers.

The (artificial) neurons and their synapses are called Perceptron. Each neuron performs the weighted average of the signals received at the input, with the weights are stored in the synapses and the result of this weighting is used as input for activation function [Cor02]. Finally, the output of this activation function is response (result) of the neuron (Illustration 4).

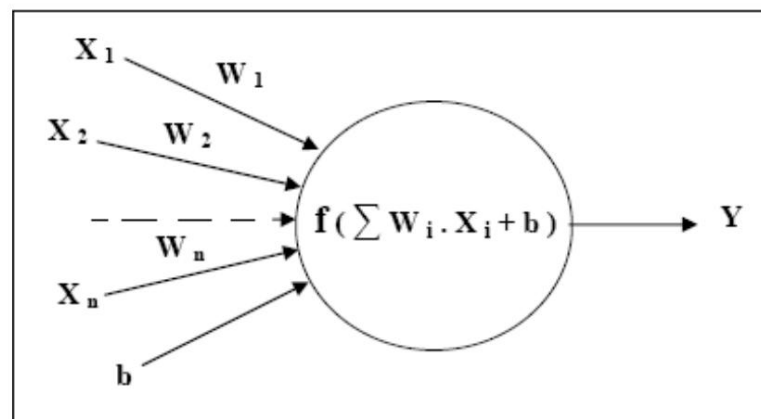


Illustration 4 - Artificial neuron of a Perceptron (Excerpt [BCL07]).

The ability of Artificial Neural Networks to learn and generalize learned information, to act as universal function mappers multivariate, with computational cost that grows only linearly with the number of variables [BCL07] were the reasons for its choice in the mining problem PBEM data.

¹¹ the

¹¹ The ability to generalize, according to Haykin (1999), refers to the ability of the Artificial Neural Network to produce outputs suitable for inputs that were not present during training.

4.2 Learning Process

An important feature of ANNs is their ability to learn through of samples. The procedure used to carry out the learning process is called a learning algorithm, this function is to modify the weights synaptic network in an orderly fashion to achieve a desired goal [Hay99], or that is, it seeks the best network performance according to established criteria previously.

Learning is defined as a process by which the free parameters of a neural network are adjusted through a continuous form of stimulation by the external environment , with the specific type of learning being defined by the way particular how the adjustments of the free parameters occur [MM70].

This definition of the learning process implies the following sequence of events [Hay99]:

1. The neural network is stimulated by an environment.
2. The neural network undergoes modifications in its free parameters as a result of this simulation.
3. The neural network responds in a new way to the environment, due to the changes that occurred in its internal structure.

There are a variety of neural network learning algorithms, each each offers its own advantages. Learning algorithms differ from each other in the way in which adaptation to a neuron's synaptic weight is formulated. Another factor to be considered is the way in which a neural network, composed of a set of interconnected neurons, concerns its environment.

Five learning rules are known [Hay99], namely:

- Learning by error correction;
- Memory -based learning ;
- Hebbian learning;

- Competitive learning and
- Boltzman learning .

All are used for the design process of neural networks. You can divide them basically in the form of their learning, such as supervised or unsupervised. supervised which concerns the use or not of a teacher . “ ”

In the study of supervised learning, an essential provision is a "teacher" capable of providing exact corrections to the network, stimulating its inputs through input patterns and observing the output calculated by the same comparing with the desired output. The network response being a function of the current values of the weights, these are then adjusted to bring the network output closer to the output that is wants [Hay99]. The reduction of the difference between the network output and the expected output is carried out through small adjustments in the weights at each stage of training (Illustration 5).

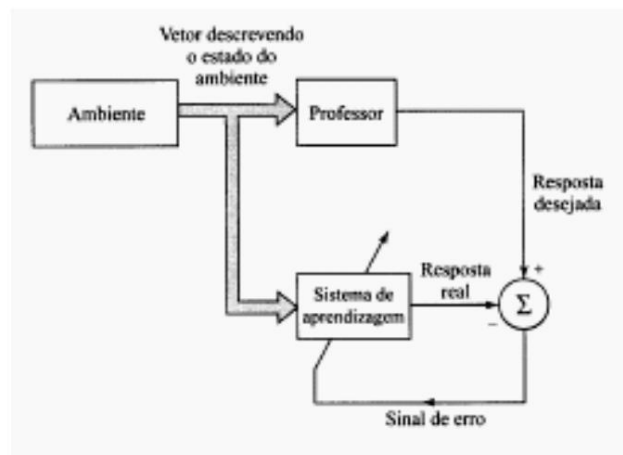


Figure 5 - Supervised learning scheme (Excerpt [Hay99]).

In the study of unsupervised learning, there is no teacher (supervisor) for monitoring and corrections in the learning process, that is, there are no labeled examples of the function to be learned by the neural network. “ ”

Unsupervised learning can be divided into:

- Reinforcement learning: learning is accomplished through intense interaction with the environment, with the aim of minimizing a scalar index performance. In reinforcement learning, a critic converts a primary reinforcement signal received from the environment in reinforcement signal with better quality (heuristic signal) (Illustration 6).

Advantage of reinforcement learning – interaction with the environment, developing the ability to learn, using only the results of your experience.

Disadvantages of reinforcement learning there is no teacher who provide a desired response at each step of the learning process, which would optimize the result of the training stages . Another disadvantage is that the delay in generation of the primary reinforcement signal, it follows that the learning machine must decide individually on each action in the sequence of time steps that lead to the final result [Hay99].

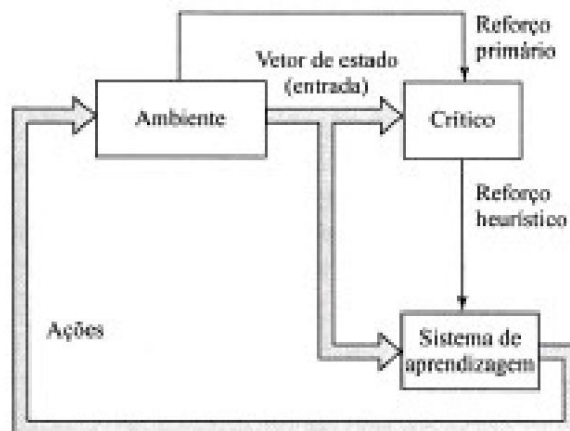


Figure 6 - Reinforcement learning scheme (Excerpt [Hay99]).

- Unsupervised (self-organized) learning : in this type of learning, there is no critic to supervise learning as in reinforcement learning. The free parameters of the network are optimized against a task-independent measure defined

for learning the network. This develops skills to form internal representations (Illustration 7).



Figure 7 - Unsupervised learning scheme (Excerpt [Hay99]).

The most common way of using ANNs is learning through a dataset [BCL07]. Some basic learning tasks for ANNs are [Hay99]:

- Pattern Association: a task that takes two forms:
 - o Self-association, where a neural network stores a set of patterns that are repeatedly presented to the network. Soon after then a distorted description of what was presented is presented originally stored. The task is to find the pattern particular. In this form, non-commercial learning is used. supervised.
 - o Heteroassociation, where an arbitrary set of patterns of input is associated with another arbitrary set of patterns output. In this form, supervised learning is used.
- Pattern Recognition: process by which a received pattern is assigned to one class from a predetermined number of classes.
- Function Approximation: process by which a pattern is presented to the inputs and outputs of the network and for each presentation the weights are adapted to map input and output relationships. In this task supervised learning is used.

- Control: process by which a condition is maintained under control. important part of a system.
- Filtering: process that uses an algorithm with the function of *extracting information about a particular quantity of interest from a noisy dataset* ”.
- Beamforming: process used to *distinguish between “ spatial properties of a target signal and the background noise”*.

4.3 Single Layer Perceptron Networks

It was in 1958, through the work of Frank Rosenblatt, that the concept of learning in artificial neural networks. This proposed model was composed of a network structure having basic units of neurons and by a rule of learning[Ros58]. It became known as the perceptron. Rosenblatt demonstrated that a *McCulloch and Pitts (MCP)*¹² neuron trained with the learning algorithm of the perceptron, always converges if the problem in question is linearly separable

[Ros58]. Rosenblatt's work has been discredited, as to its computational capacity of the model, because of the criticisms made by Minsky and Papert in the 70s [MiP69]. In 1982, Hopfield published work on neural networks and the back-propagation, which led to new work using neural networks emerging with new impetus.

The perceptron, when using a neuron, can only perform classification of patterns with only two classes. If the output layer of the perceptron to include more than one neuron, classification of more than one can be performed. two classes, however, the classes must be linearly separable (Illustration 8) to

¹² MCP Warren McCulloch and Walter Pitts, in 1943, were the pioneers in proposing a model artificial of a biological neuron.

for the perceptron to work correctly [Hay99], which in *world* problems-*real*¹³ is less likely to occur.

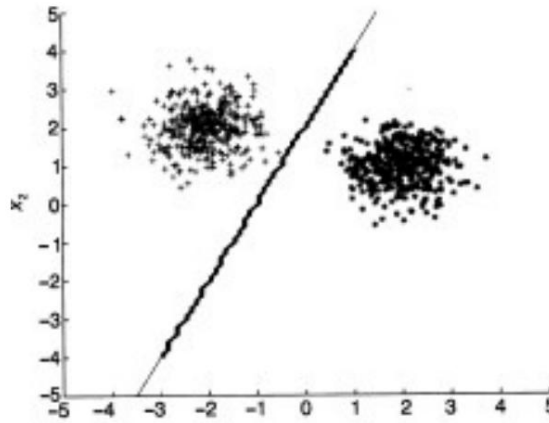


Figure 8 - Graphical representation of linearly separable classes (Excerpt [Hay99]).

4.4 Multilayer Perceptron Networks

Multilayer neural networks, with intermediate layers formed by neurons with a sigmoidal activation function¹⁴ are called *Multilayer Perceptron* (MLP). Unlike single-layer networks, they are capable of solve problems of non-linear characteristics, through activation functions of each neuron in the network and the composition of its structure in successive layers [BCL07]. Single intermediate (hidden) layer MLPs are sufficient for approximate any continuous function [Cyb89].

The layers in an MLP network are arranged in sequence, one after the other. The Figure 9 shows a model of a typical MPL network with an intermediate layer. The processing performed by each neuron in a given layer is defined by combining the processing carried out by the neurons in the previous layer that are connected to them.

¹³ Real-world problem: concerns real-world situations that can be seen as problems or opportunities.

¹⁴ The most frequent activation function in MLP networks is the sigmoid function, given by: $y = 1 / (1 + \exp(x))$.

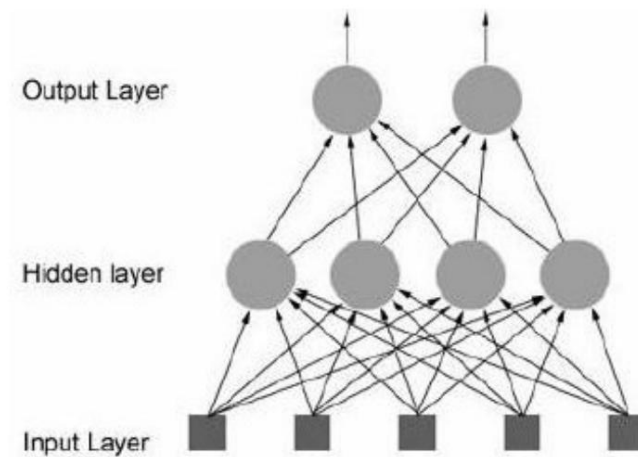


Figure 9 - MLP network with an intermediate layer (Excerpt [Hay99])

The most widely used algorithm for training ANNs is Backpropagation [Cru07]. Because it is supervised, it uses input and output pairs (x, y_d) to, through of an error correction mechanism, adjust the network weights. Training occurs in two phases, with each phase traveling through the network in one direction. These two phases are called *forward* and *backward* phase. The *forward* phase is used to define the output of the network for a given input pattern. The *backward* phase uses the desired output and the output provided by the network to update the weights of its connections [BCL07]. That is, in the *Backpropagation* algorithm, in each learning cycle, the errors obtained by the differences between the network outputs and the training values are propagated in the direction opposite (backwards), after which the weights of the connections are adjusted. The training usually ends when the minimum error is obtained at the output, in the case of regression, or the minimum misclassifications [Cru07].

The *Backpropagation* algorithm allows the automatic adjustment of the weights of synapses.

The calculation of the error for the output of a neuron is given by:

$$e_i(k) = y_i(k) - d_i(k)$$

where:

$i \rightarrow$ neuron

$e_i \rightarrow$ error of neuron i

y_i -> output of neuron i

d_i -> desired output of neuron i

k -> input in question

The total error is:

$$e(k) = \sum_{i=1}^N \frac{1}{2} e_i^2(k)$$

where N is the number of neurons.

After calculating the error, the synapse weights are updated [Fah98]:

$$\Delta w_{ij} = w_{ij}(k+1) - w_{ij}(k) = -\eta \frac{\partial e(k)}{\partial w_{ij}}$$

where:

w_{ij} -> weight of synapse ij ;

η -> learning control (from 0 to 1)

The weights are then adjusted according to the formula:

$$w_{ij}(k+1) = w_{ij}(k) - \eta \cdot y_i(k) \cdot e_j(k),$$

where k indicates the current iteration of the algorithm. That is, the weight of a synapse w_{ij} will be result of the difference between the previous weight and the product of the control factor of learning by the output of neuron i (beginning of the synapse) and by the error of neuron j (end of the synapse). The learning control factor must increase or decrease the effect of the error in synapse weight, which can lead to faster or more rapid convergence delayed, depending on whether its value is closer to 1 or 0 [Cru07]. Then we have two situations:

- If convergence is faster, it may fall into a local minimum or never converge due to the variation of weights.

- A slower search may lead to the global minimum, but will require quite a while.

Choosing this value is essential, as it will allow direct control of capacity. network generalization [ZuG93]. A new term is added for greater control of process; the term momentum, where it generates a new weight by the formula:

$$w_{ij}(k+1) = w_{ij}(k) - \eta \cdot y_i(k) \cdot e_j(k) + \mu \Delta w_{ij}(k-1)$$

The difference lies in the introduction of a term that corresponds to the adjustment of the previous measured iteration of a factor μ , called inertia (momentum), which helps convergence to a global minimum, as it allows part of the iteration adjustment previous will be reflected in the current setting [Cru07].

An important point to note is the number of iterations to be performed. through the network for learning. An excessive number of iterations can lead to *overfitting*¹⁵.

A balance must be struck between precision and generalizability to stop of training, through criteria such as [ZuG93]:

- Maximum error;
- Error gradient ;
- Number of iterations;
- Cross-validation,

¹⁵ Situation in which the ANN adapts very well to learning cases, but will respond poorly to others.

5 PBEM Data Mining

Business Understanding, Understanding and Data Preparation

This chapter covers the first three steps of the methodology selected for the PBEM data mining project.

5.1 Introduction

The methodology used in this work was CRISP-DM (*Cross-Industry Standard Process for Data Mining*). Although this has a disadvantage in terms of respect for the reuse of knowledge in relation to the DMLC methodology¹⁶ [WiH00] as shown in Table 2 [Cun05] with a summary of the main methodologies for data mining projects, for the Bolsa Program mining work. The school opted for the CRISP-DM methodology due to the reuse of knowledge, in this case, it is not relevant and because it is the most widespread methodology in World17.

Ano	Metodologia	Iterativo	Interativo	Organizado em Fases	Processos Extras	Reuso de Conhecimento	Visão Projeto	Levantamento Requisitos	Entendimento Negócio
1996	Fayyad	A	M	A	N	N	N	N	N
1996	CRISP-DM	A	M	A	N	N	A	M	A
1996	Brachman	A	M	A	M	N	N	M	M
1997	Klemettine	A	M	A	N	N	N	N	A
1998	Feldens	A	M	A	N	N	N	N	N
2003	DMLC	A	M	A	N	M	A	M	A

N= não aborda; M= menciona, mas sem detalhes; A= aborda em detalhes.

Table 2 - Comparison of the main methodologies for data mining projects (Taken from [Cun05]).

¹⁶ DMLC (Data Mining Life Cycle) is a methodology for data mining projects, proposed with the aim of bringing improvements to the CRISP-DM methodology with regard to the reuse of knowledge and documentation of the resources involved.

¹⁷ Most widely used methodology in the world: CRISP-DM - 51%. Source: KDnuggets research conducted in July 2002. Available at <http://www.kdnuggets.com/polls/2002/methodology.htm>. Accessed on March 2009.

CRISP -DM is a non-proprietary methodology developed in 1996 by an international consortium of companies, which is structured around tasks and objectives for each phase of a data mining project [Cri96].

The methodology consists of 6 phases (Illustration 10):

Business understanding: at this stage a clear vision of the business is developed. needs to be satisfied;

Understanding the data: at this stage the available data is identified and where they are located;

Data preparation: at this stage the data must be prepared according to the method to be used in the modeling stage;

Modeling: at this stage, artificial intelligence algorithms are used to problem to be treated.

Evaluation: at this stage the result found is evaluated by comparing it with the objective of the mining project.

Distribution of results: in this phase the project results are made available to those interested.

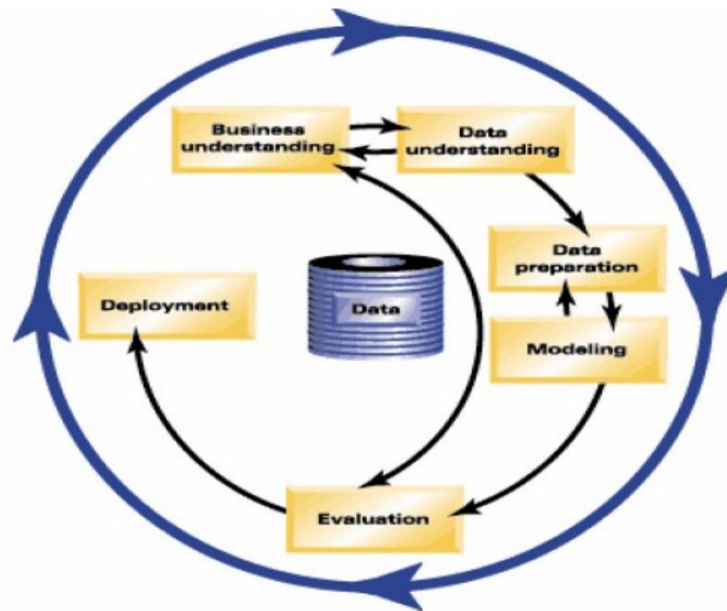


Illustration 10 - Phases of the CRISP-DM Methodology Processes.

The phases of understanding the business, understanding and preparing the data and, their applications to PBEM data mining are described in the following sections. The modeling and evaluation phases are in Chapter 6. The section on understanding the data had its content reduced to avoid redundancies, since it was the subject of the Chapter 2. The distribution phase is not part of the scope of this work, being limited to documentary delivery to the expert in the field.

5.2 Business Understanding

To understand the business - functioning of PBEM - the following were carried out: activities discussed below:

- Four visits to the domain expert;
- Exchange of e-mails for questions and clarifications regarding the problem;

- Monitoring visits carried out by the Bolsa Program team
School for families benefiting from the Program or who may be benefited;
- Defining PBEM success.

Highlight for the penultimate activity in which two visits were carried out monitoring of the beneficiary and three visits for possible inclusion of new beneficiaries.

By observing the facts *on site*, talking to the applicants for the benefit and with the experts, it was possible to confirm that the PBEM benefit is used basically for 3 different situations:

- To ensure a minimum situation suitable for keeping the child in school;
- For the survival of the family. An example is in images, taken during one of the visits, of a house where six people live (one adult, one teenager and four children) (Appendix B).
- In an improper manner.

As the objective of this work is to improve the decision-making process for granting of the benefit through analysis of data from appropriate claimants (claimants who currently receiving the benefit) and those on leave (applicants who at some point received the benefit, but were removed for various reasons), the initial amount of records (instances) to be analyzed is 10,770 (ten thousand, seven hundred and seventy).

Conversations with the PBEM domain expert led to the definition of a new target class identifying the Program's success stories. According to the specialist in the PBEM domain the success of the program is recognized as the continuity of the applicant in the Program, excluding any absences other than those separations considered natural, which are the arrival of the age of fourteen and twelve months of the dependent and obtaining a per capita income greater than 1/3 of the minimum wage.

Therefore, for this work it was necessary to identify the cause of the absence to inform RNA which are the good and bad participants in the Program. For this reason the target variable TYPE was created as will be detailed in section 5.4.2.

The operating point of the proposed system is shown in Illustration 11. The point of operation was chosen observing the non-existence of a posteriori data (for This work concerns the next activity after confirmation of the data of applicants) and as a tool to support the decision-making process of the evaluation committee.

5.3 Understanding the Data

5.3.1 Data Source

To understand the data, the following activities were carried out:

- Seven appointments for visits to data experts, however only three successful (realized);
- Exchange of e-mails for questions and clarifications regarding data;
- Exploratory data research.

For this last task, it is worth highlighting the difficulty in finding answers through database search, since there are many tables (entities) inconsistent (they stopped being updated at a given time; they are updated, but do not portray the reality of the facts or were never used), the descriptions of the fields do not allow an understanding of their content and finally the lack of a data dictionary for all entities (only exists for the tbRequerente entity).

The data was obtained in September 2008 and contains the record of all families registered in the PBEM until then-60,451 (sixty thousand, four hundred and fifty-one one records) families and takes up 728 MB of space.

The Bolsa Escola database is made up of 184 related tables, being the main ones are tbApplicant (has applicant data) and tbDependent (has

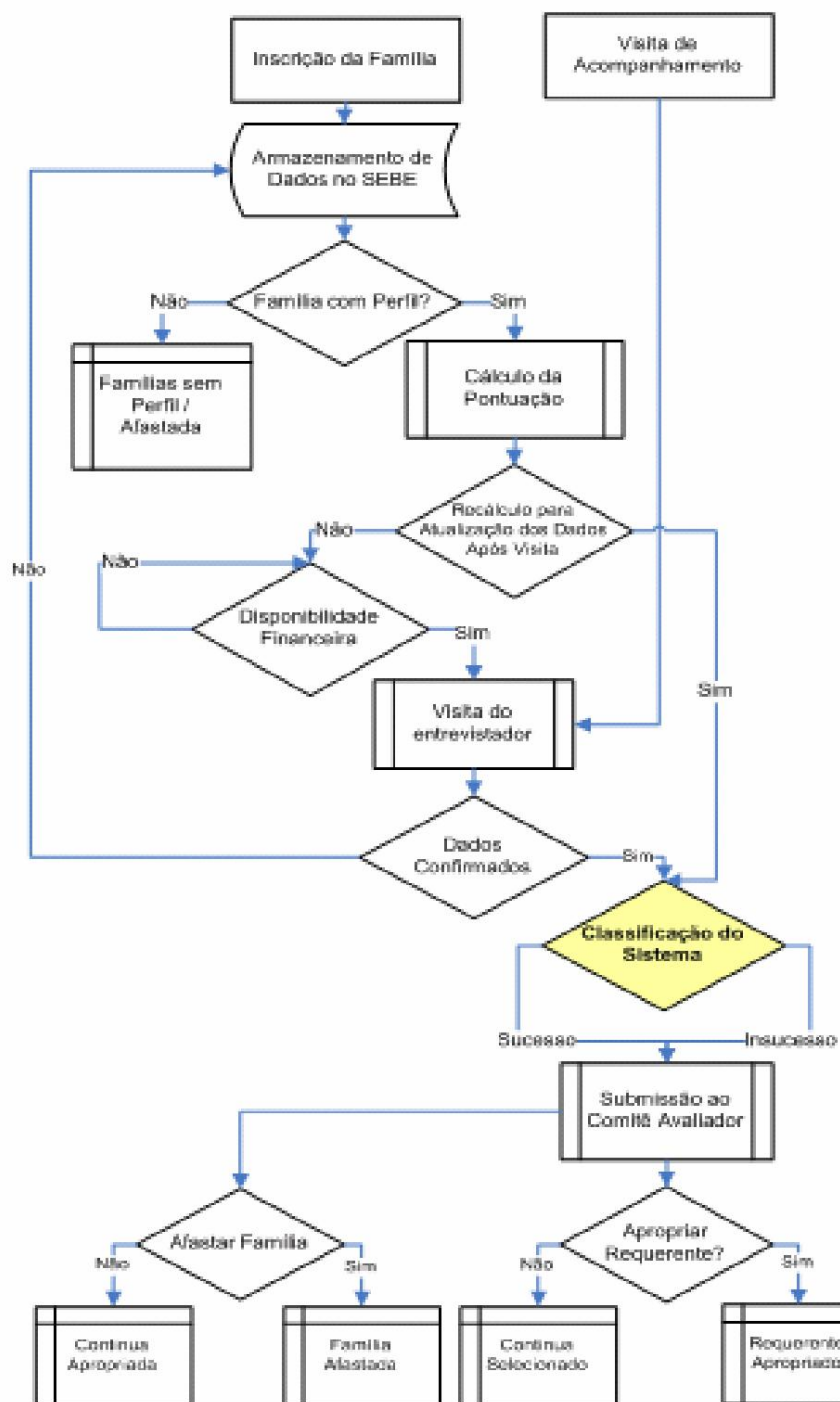


Illustration 11 - Identification of the operating point of the proposed system.

data of each applicant's dependents). The first of these is the most important in this work that has a level of granularity in the applicant . The entity-model database relationship is in Appendix C.

5.3.2 Applicant Registration Database

This database stores the registration data of PBEM applicants provided in the moment of registration for the Program at the HEADQUARTERS of the City Hall of Recife. This entity has 102 attributes that store various information about the applicant, such as name (nrequenome), sex (crequesexo), date of birth (drequenasc), street (erequeende), spouse 's data (nrequencon, crequescon, drequednco, cestadufco, drequechco, crequeidco, crequeorco, casodufic, drequeemic, cgrauicodc, ccbotbcodc, ccbotbcoc2, csittrcodc, drequedesc, frequesinc and csitescodc) and social economic data (drequecheg, cescivcodi, cgrauicodi, ccbotbcodi, ccbotbcodo, vrequerend, vrequeperc, crequeener, crequeagua, crequesani, arequenuco, arequememb, frequebfam and others). tbRequerente entity data dictionary is in Appendix D.

A survey was carried out for the target class on the completion of main fields highlighted by the domain expert that influence the choice of the beneficiary.

Filling in data regarding the existence of electrical energy:

Crequeener (Electric Energy)		
Category	Qty.	%
Yes	96	1.112
No	52	4%
Total	1,164	100%

Table 3 - Completion of data regarding the existence of electrical energy.

Filling in data regarding the existence of running water:

Crequeagua (Pipe Water)		
Category	Qty.	%
Yes	959	82%
No	205	18%
Total	1,164	100%

Table 4 - Completion of data regarding the existence of piped water.

Filling in data regarding the existence of basic sanitation:

Crequesani (Basic Sanitation)		
Category	Qty.	%
Yes	876	75%
No	288	25%
Total	1,164	100%

Table 5 - Completion of data regarding the existence of basic sanitation.

Filling in the data regarding the number of rooms in the house:

Arequenuco (Number of Rooms)		
Category	Qty.	%
1	162	14%
2	216	19%
3	182	16%
4 or more	604	52%
Total	1,164	100%

Table 6 - Completing the data regarding the number of rooms in the house.

Filling in the data regarding the number of family members:

Arequememb (Number of Family Members)		
Category	Qty.	%
Up to 3	175	15%
4 to 7	797	68%
8 to 11	174	15%
12 or more	18	2%
Total	1,164	100%

Table 7 - Completion of data regarding the number of family members.

Filling in the data regarding the family income value:

Family Income Value		
Category	Qty.	%
Up to	612	53%
207.5 to 207.51	443	38%
to 415.1 or more	109	9%
Total	1,164	100%

Table 8 - Completing data regarding family income.

Filling in data regarding the family's per capita income:

Vrequeperc (Value of Income Per Capita)		
Category	Qty.	%
Up to 100	1,136	98%
101 to 200	26	2%
201 or more	2	0%
Total	1,164	100%

Table 9 - Completing data regarding the family's per capita income.

Filling in data regarding the benefit receipt indicator

Federal Government Family Allowance:

Frequerfam (Family Allowance Indicator)		
Category Qty.		%
Receive	960	82%
Does not receive	204	18%
Total	1,164	100%

Table 10 - Completion of data regarding the Bolsa Família benefit receipt indicator .

When considering the entire base of beneficiary and former beneficiary applicants by the Program (Table 11), a percentage much lower than expected is observed , a since the Federal Government's Bolsa-Família Program covers approximately 100,000 (one hundred thousand) families¹⁸ and the socio-economic profile of the participants of both Programs are similar. The application domain expert agrees with this fact and believes that there may be inconsistency in the database.

Frequerfam (Family Allowance Indicator)		
Category Qty.	%	
Receive	2,588	24%
Does not receive	8,182	76%
Total	10,770	100%

Table 11 - Completion of data regarding the Bolsa Família benefit receipt indicator for beneficiaries and former beneficiaries.

Filling in the data regarding the applicant's marital status:

Cescivcodi (Civil Status of the Applicant)		
Category Qty.		%
Married 136		12%
Divorced	2	0%
Separate	102	9%
Single	500	43%
Simple Union	344	30%
Widower	80	7%
Total	1,164	100%

Table 12 - Completion of data regarding the applicant's marital status.

¹⁸

Filling in the data regarding the applicant's level of education:

Cgraucodi (Applicant's Level of Education)		
Category	Qty.	%
Illiterate	236	20%
Incomplete Fundamental Teaching	126	11%
Compl. Fundamental Ens.	704	60%
Incomplete Secondary Education	55	5%
High School Compl.	40	3%
Incomplete Higher Education	1	0%
Higher Compl.	2	0%
Not informed	-	0%
Total	1,164	100%

Table 13 - Completion of data regarding the applicant's level of education.

Filling in the data regarding the type of occupation of the property:

Ccbotbcodo (Housing Occupation)		
Category	Qty.	%
RENTED	226	19%
RENTED	2	0%
GRANTED	153	13%
FINANCED	1	0%
INVASION	238	20%
IT DOES NOT HAVE	1	0%
OTHER	1	0%
OWN	542	47%
Total	1,164	100%

Table 14 - Completion of data regarding the type of occupation of the home.

Filling in the data regarding the type of house fence:

Ctpvedcodi (Sealing Type)		
Category	Qty.	%
ADOBE	5	0%
MASONRY	910	78%
WOOD	155	13%

WOOD	41	4%
MATERIALUSED	1	0%
IT DOES NOT HAVE	1	0%
OTHER	1	0%
TAIPANAORECESTED	44	4%
COATED TAIPARE	6	1%
Total	1,164	100%

Table 15 - Filling in the data regarding the type of house fence.

Filling in the data regarding the type of floor in the house:

Ctppiscodi (Floor Type)		
Category	Qty.	%
CERAMICS 68		6%
SUBFLOOR	361	0.31%
STILTS	1	0%
TERRABATIDA	190	16%
BRICK/CEMENT	544	47%
Total	1,164	100%

Table 16 - Filling in the data regarding the type of flooring in the house.

Filling in the data regarding the type of roof of the house:

Ctpcobcodi (Ceiling Type)		
Category	Qty.	%
SLAB	242	21%
IT DOES NOT HAVE	5	0%
PLASTIC/CARNABY	169	15%
CERAMICS TILES 509		44%
ZINC/BRAZILIT	239	21%
Total	1,164	100%

Table 17 - Completing data regarding the type of roof of the house.

Filling in the data regarding the type of land:

Cgletecodi (Gleba da Terra - Portion of land)		
Category	Qty.	%
TENANT	175	15%
TENANT	270	23%
PROP. RELATIVE	163	14%
PROP. BOSS	30	3%
OWNER	526	45%
Total	1,164	100%

Table 18 - Completion of data regarding the type of land used for the residence.

Filling in the data regarding the type of construction of the house:

Ctpconcodi (Construction Type)		
Category	Qty.	%
IN FINISHING 322		28%
STARTED	56	5%
OTHERS	69	6%
PARALYZED	283	24%
READY	345	30%
NO INFORMATION	89	8%
Total	1,164	100%

Table 19 - Completion of data regarding the type of housing construction.

Filling in the data regarding the applicant's age:

Crequeidad (Age of the Applicant)		
Category	Qty.	%
UP TO 30 YEARS 45 31 TO		4%
50 YEARS 51 TO	838	72%
70 YEARS	255	22%
71 OR MORE	26	2%
Total	1,164	100%

Table 20 - Completion of data regarding the applicant's age.

Attribute created from attribute containing the date of birth of the applicant (section 5.4.2).

Filling in data regarding the age of dependents:

Qreque0A15 (Number of Applicants who have Dependents up to 15 years old)		
Category	Qty.	%
0	149	13%
1	400	34%
2	315	27%
3 or more	300	26%
Total	1,164	100%

Table 21 - Completion of data regarding the age of dependents for applicants who have dependents up to 15 years old.

Attribute created (section 5.4.2) from attribute containing date of birth of the dependent in the tbDependente entity.

Filling in data regarding the age of dependents:

QrequeAp16 (Number of Applicants who Have Dependents as of 16 years)		
Category	Qty.	%
0	203	17%
1	278	24%
2	288	25%
3 or more	395	34%
Total	1,164	100%

Table 22 - Completion of data regarding the age of dependents for applicants who have dependents aged 16 or over .

Attribute created (section 5.4.2) from attribute containing date of birth of the dependent in the tbDependente entity.

Filling in data regarding the age of dependents:

nrequelddc (Age of the Applicant's Conj.)		
Category Qty.		%
UP TO 15	1.015	51.36%
YEARS FROM 16 YEARS	961	48.63%
Total	1.976	100%

Table 23 Number of dependents by age group.

Filling in data regarding the age of the applicants' spouses :

nrequelddc (Age of the Applicant's Conj.)		
Category Qty.		%
UP TO 30 YEARS	20	2%
31 TO 50 YEARS	1.021	88%
51 TO 70 YEARS	110	9%
71 OR MORE	13	1%
Total	1,164	100%

Table 24 - Completion of data regarding the age of the applicants' spouses.

Filling in data regarding the applicant's gender:

Crequesexo (Sex of the Applicant)		
Category Qty. %		
F 997 86%		
M	167	14%
Total	1,164	100%

Table 25 - Filling in data regarding the applicant's sex

5.4 Data Preparation

Because there is no Data Warehouse for the Bolsa Escola database and therefore routines for proper extraction, processing and loading of data, a high level of

inconsistencies and outliers in the database were found, which took a lot of time to preparing data for modeling.

The initial task was to group all the applicant's socio-economic data into a single denormalized structure that now contains 128 columns. For this purpose, following activities were performed:

- Replacement in the tbRequente table of fields with codes by respective descriptions; in a data warehouse it would be to bring the descriptions from dimension tables to the fact table.
- Standardization of descriptions for each field, removing typing errors and unnecessary variations. Inconsistency treatment was performed assigning the mode of each attribute. The problem of missing data was solved by calculating the average value for each numeric attribute of each class and replacing missing values with this mean value calculated.
- Creation of fields with information in order to make the process more efficient mining result;
- Elimination of less significant variables;
- Preparing variables for the mining algorithm to be used for modeling.

5.4.1 Data Filtering and Cleaning

As mentioned in section 5.2, for the purpose of the work, 10,770 records (instances) would be used in data mining.

Initially, eight records were deleted as part of data cleansing.

The reason is due to data inconsistency, as all the codes of the applicants table tbRequerente must be included in the table tbHistoricalDependent which contains the reasons for the absences, so the database had 10,762 records. This observation was communicated to the data experts .

As for the columns (attributes), the first step was to eliminate the columns with direct relationship (code and description) and those that uniquely identified the record in the table [CCK00]. Below is a list of the columns removed and their respective reasons:

Column	Description	Reason for Withdrawal
CESCOLCODI School Code		It was removed, since the analysis by neighborhoods was chosen.
CBAIRRCODI Neighborhood Code		A field was left with a description of the Neighborhoods
CNACIOCODI	Code of Nationality	It was removed due to high incidence of a single class.
CPROPFCODI	Phone Owner Code	It was removed due to the low level of reliability of the information, as discussed with the domain expert.
CESCIVCODI	State Code Applicant's Civil	Field was left with description of marital status
CGRAUICODI	Degree Code Instruction of the Applicant	A field was left with a description of the level of education
CCBOTBCODI Profession Code		A field was left with a description of the profession
CCBOTBCODO Occupation Code		A field was left with a description of the type of occupation
CSITTRCODI	Employment Status Code	A field was left with a description of the work situation
CSITESCODI	Situation Code Special	It was removed due to low occurrence (1% of the base)
CGRAUICODC	Degree Code Instruction of the Applicant's spouse	A field was left describing the applicant's spouse's level of education.
CCBOTBCODC	Spouse's Occupation Code	A field was left with a description of the spouse's profession
CCBOTBCOC2	Spouse's Occupation Code	A field was left with a description of the spouse's occupation
CSITTRCODC	Situation Code in Spouse's Labor Market	It was removed due to redundancy with the description of the occupation
CSITESCODC	Situation Code Spouse Special	It was removed due to low occurrence (1.8% of the base)
COCUMOCODI	Housing Occupation Code	A field was left describing the type of occupation of the house
CTPVEDCODI	Type Code Sealing	A field was left with a description of the type of sealing
CTPPISCODI	Type Code Floor	Field was left with description of the type of floor
CTPCOBCODI	Type Code Coverage	Field was left with description of the type of coverage
CGLETECODI	Gleba Code Earth	It was removed due to redundancy with the description of the type of occupation

CTPCONCODI	Type Code Construction	A field was left with a description of the type of construction
CSTATUCODI	Status Code Applicant	It was removed because it contained a posteriori data.
CORIGECODI	Registration origin code	It was removed , since, according to the expert in the field, it is not relevant
AREQUECTPS	Wallet Number Professional	Not relevant to the problem
AREQUEFOLH	Sheet Number CN	Not relevant to the problem
AREQUEPHONE	Contact Phone Number	Not relevant to the problem
AREQUEISOC	Registration Number Social (PIS/PASEP/SUS)	Not relevant to the problem
AREQUENCPCF	CPF Number	It was removed because it only identified one record in the table.
AREQUENLIV	Book Number CN	Not relevant to the problem
AREQUESECP	CP Series	Not relevant to the problem
AREQUETECN	Term Number Certificate of Birth	Not relevant to the problem
CESTADCOCP	No Information	It was removed since 100% of the records are PE and N/A(not evaluated - no filled)
CESTADCDC	CN Issuing State	Not relevant to the problem
CESTADCODI	Issuing State of CI	It was removed , since 98% of the records are from PE and N/A (not evaluated - not filled)
CESTADCODM	Housing State	It was removed , since 100% of the records are from PE
CMUNICEO	Code of the Municipality of Origin	Non-standard filling
CMUNICEP	Code of the Municipality of Residence	Records , since 100% of the were removed from Recife
crequecodi	Applicant Code	It was removed because it only identified one record in the table.
CREQUEDDDF	Telephone Area Code	It was removed, since 100% of the records are PE and N/A (not evaluated - not filled)
CREQUEIDCO	Driver's License Number Spouse Identity	It was removed because it was a unique code, and 68% were N/A.
CREQUEINSC	Ancient Inscription of Applicant	It was removed because it only identified one record in the table.
CREQUENUID	Driver's License Number Identity	It was removed because it identified only one record in the table
CREQUENUME	House Number Non-standardized field	
CREQUEPONT	Score	It was removed because it contained a posteriori data.
CRACKETS	School Type (E- State, M- Municipal)	It was removed , since 100% of them are municipal schools in Recife

CREQUETYPE		It was removed due to the high number of non-completions (60%)
CREQUEZONA	Residential Area	It was removed, since 100% of the data removed belong to the same class
CTPDOCCODI		It was removed, since 100% of the data removed belong to the same class
DREQUEANFA	Year of Manufacture Vehicle	It was removed, since 100% of the data belong to the same class
DREQUEAPRO	Date used for internal system procedure	Not relevant to the problem
DREQUECHCO	Arrival Date of Spouse	It was removed, since 100% of the data belong to the same class
DREQUECHCO	Arrival Date	Not relevant to the problem
DREQUECTPS	Issue Date	It was removed, since 100% of the data removed belong to the same class
DREQUEDESC	Spouse's Unemployment Date	It was removed, since 100% of the data removed belong to the same class
DREQUEDESE	Unemployment Date	It was removed, since 100% of the data removed belong to the same class
DREQUEDNCO	Date of birth of the Spouse	It was removed, however, before it was used to generate a new variable - spouse's age
DREQUEEMCN	Issue Date CN	Not relevant to the problem
DREQUEEMIC	Ci Issue Date of the Spouse	Not relevant to the problem
DREQUEEMII	Date of Issue of the CI	It was removed, since 100% of the data removed belong to the same class
DREQUEEXCE		It was removed, since 100% of the data belong to the same class
DREQUEEXPO	Date used for internal system procedure	
DREQUEINPG		It was removed, since 100% of the data removed belong to the same class
DREQUEINSC	Registration Date	Not relevant to the problem
DREQUENASC	Date of Birth	It was removed, however, before it was used to generate a new variable - applicant's age
DREQUEPONT	Date used for internal system procedure	Not relevant to the problem
DREQUESELE	Date of applicant selection	Not relevant to the problem
DREQUETRAN		It was removed, since 100% of the data removed belong to the same class
EREQUECOMP	Complement of the address	Non-standard filling
EREQUEENDE	Street	Non-standard filling
EREQUEINFC	Information Complementary to the Applicant	Non-standard filling
EREQUENCEP	Zip Code Number	We opted for analysis by neighborhoods

EREQUEREFE	Reference Point	Non-standard filling
ESITESDSCC	Description of Spouse's Special Situation	It was , since 100% of the data removed belong to the same class
ESITTRDSCC	Description of the Work Situation of Spouse	It was removed , since 80% of the data not filled.
STATUDESC	Description of the Applicant's Situation in Program	It was removed because it contained a posteriori data.
FREQUEPONT	Flag for applicants scored in the Program	It was , since 100% of the data removed belong to the same class
FREQUEVEIC	Vehicle Indicator (C/M)	It was removed , since 100% of the data belong to the same class
FREQUEVISI	Interviewer 's Visit Flag	Unreliable data
reasonRemove	Reason for removal of the applicant	It was removed, however, before it was used to generate a new variable - type of removal
NREQUECART	Name of the Registry Office CN	Not relevant to the problem
NREQUEENTR	Name of the Interviewer	Not relevant to the problem
NREQUENCON	Spouse's Name	Not relevant to the problem
NREQUENOME	Applicant's Name	Not relevant to the problem
NREQUENOMM	Mother's Name	Not relevant to the problem
NREQUENOMP	Father's Name	Not relevant to the problem
TRICKLE	Date used for internal system procedure	Not relevant to the problem
VREQUEVAVE	Percentage Income Value captures	It was removed , since 100% of the data not filled.

Table 26 - List of columns removed from the base and their respective reasons.

5.4.2 Creating Fields

New attributes were created during the data aggregation process, with objective of gaining information for modeling in the ANN.

> CREQUEIDAD: Applicant's age. Created from the attribute DREQUENASC (date of birth).

> QREQUEQTDP: Number of Applicant's Dependents.

> QREQUE0A15: Number of Dependents up to 15 years of age

> QREQUEAP16: Number of Dependents from 16 years of age

> NREQUEIDDC: Applicant's Spouse Age

> TYPE: Type of applicant. Target Variable. Created with the aim of marking the applicant as Success or Failure (target class), as defined by domain expert. 89.18% of the instances were classified as Success and “ ” 10.82% as Failure in PBEM.

5.4.3 Selection of Most Relevant Variables

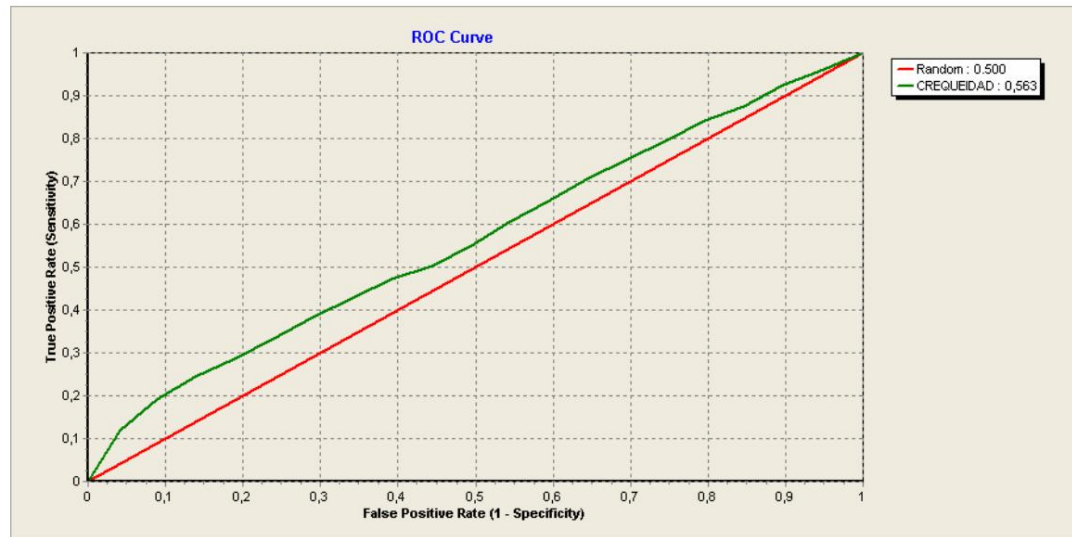
After eliminating eighty- seven variables and creating six others, with aim to confirm the relevance of all forty-six variables (attributes) remaining (sixteen numerical and thirty categorical), techniques were used to evaluate the importance of each one. All techniques had the target variable as a reference. The three techniques used, in addition to the tacit knowledge¹⁹ of the specialist PBEM, are described in the next sessions.

- **ROC curves**

ROC (*Receiver Operating Characteristics*) charts are a useful tool useful for visualizing and evaluating classification models. They are also used to evaluate how a learning system is able to order the examples, allowing an analysis independent of the classification threshold. The analysis ROC provides a richer assessment than simply evaluating the model classification from a single measurement [RP05]. Also known as curves ROC [ProF98], represent the false positives or *fp rate* (on the abscissa axis) and true positives or *tp rate* (on the y-axis). They show the relationship of the rates of false positives (FP) and true positives (VP) by varying a threshold. This relationship predicts the behavior of classifiers independently of costs.

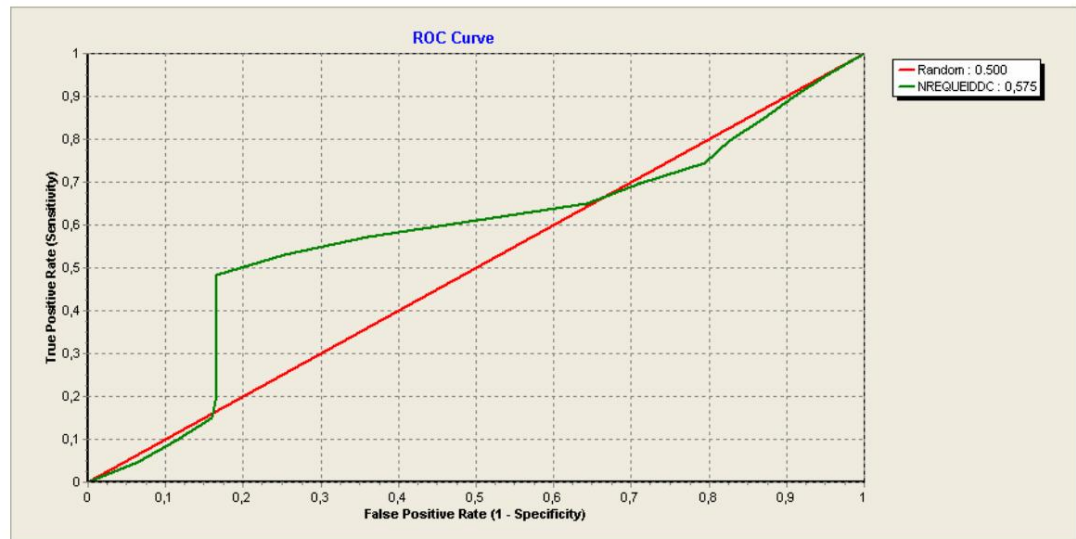
¹⁹ Tacit knowledge: subjective knowledge inherent to a person's skills.

and the distribution of classes. For each cutoff point, the sensitivity and complement of specificity ($1 - \text{specificity}$) are calculated and placed on each axis respective of a two-dimensional graph [FAW05]. ROC graphs were also used to evaluate the performance of the experimented model (section 6.2.2). Examples of two - variable ROC plots are shown in Graphs 2 and 3.



Graph 2 - ROC curve of the CREQUEIDAD variable (applicant's age).

For the continuous variable indicating the applicant's age, CREQUEIDAD, the ROC curve shows that this is a good classifier because it moves away from the line diagonal, which represents a random classifier.



Graph 3 - ROC curve of the variable NREQUEIDDC (Age of the applicant's spouse).

For the continuous variable indicating the age of the applicant's spouse, NREQUEIDDC, the ROC Curve also shows that this is a good classifier, except in $fp\ rate < 0.17$ and $fp\ rate > 0.7$ with worse classification moments (curve below the diagonal) than those of random classifiers, i.e., with little evidence make negative classifications by committing *false positive* errors .

The ranking of the main variables for the ROC curve is:

Nrequeiddc (age of the applicant's spouse) = 0.575

Crequeidad (applicant's age) = 0.563

Vrequerend (family income value) = 0.533

Arequememb (number of family members) = 0.519

QrequeA15 (number of dependents up to 15 years of age) = 0.495

QrequeAP16 (number of dependents aged 16 and over) = 0.485

- **Information Gain**

Information Gain or *Information Gain Attribute Ranking* is a technique used for variable selection that considers each variable individually, ordering them based on their predictive capabilities. In this technique, each A_i attribute (where A is an attribute of a data set) of the database is associated with a value corresponding to the information gain, in the form [Mer07]:

$$GI_i = E(C) - E(C|A_i). \text{ Where:}$$

GI – Information Gain;

$E(C)$ – Calculation of the entropy of the class attribute before observing attribute A ;

$E(C|A)$ - Calculation of the entropy of the class attribute after observing the attribute A .

Being:

$$E(C) = - \sum_{c \in C} p(c) \log_2 p(c),$$

(taken from [Mer07])

where $p(c)$ is the probability of class C occurring in the database and

$$E(C|A) = - \sum_{a \in A} p(a) \sum_{c \in C} p(c|a) \log_2 p(c|a),$$

(taken from [Mer07])

where $p(a)$ is the probability of a occurring in the database; $p(c|a)$ is the probability of class c occurs after the occurrence of the value of attribute a .

Before submitting the data to the technique, it was necessary to convert the fields categorical into binary numbers, a transformation that was also used for the phase

that precedes modeling, as described in section 5.4.5. The ranking result of the forty- five variables (the TYPE variable is the target variable) using the technique *Information Gain Attribute Ranking* is shown in Table 27 - Result of Information Gain Attribute Ranking to variables.

Order	Score	Variable	Description
1	0.022397	CREQUEIDAD Applicant's age	2 0.008351 CREQUESEXO Applicant's
3	0.004378	EORIGEDESC Description of the applicant's origin	4
0.003544	QREQUEOA15 Number of dependents up to 15 years of age.		
5	0.003209	ECBOTBDSCC Spouse's occupation	6 0.002715
CREQUESCON Spouse's gender	7 0.002712	FREQUEBFAM	
Bolsa-Familia beneficiary	8 0.002676	ECBOTBDESC Profession description	9
0.002263	EBAIRRNESC School neighborhood		
10 0.00191	EBAIRRNOME Applicant's neighborhood	11 0.001877	
ESITRDESC Description of work situation	12 0.001859	ECBOTBDSCO Description of	
occupation	13 0.001646	EGRAUIDESC Applicant's level of education	14
0.001452	AREQUEMEMB Number of family members		
15 0.001451	QREQUEAP16 Number of dependents aged 16 and over.		
16 0.001215	VREQUEREND Family income value		
17 0.001125	NREQUEIDDC Age of the applicant's spouse	Issuing	
18 0.001116	CREQUEORCO	19 0.000934	body of the spouse 's CI (identity card)
NE\$CIVNOME Description of marital status	20 0.000917	CESTADUFIC	
State issuing the spouse's CI	21 0.000917	NESTADNMIC State issuing the applicant's	
CI	22 0.000906	ESITDESC Description of special situation	23 0.000888 CREQUEORGA
CI agency	24 0.000833	NESTADNOMC State of owner	25 0.000799 EGRAUIDSCC
Level of education of spouse	26 0.000794	EPROPFDESC	
Owner of the phone	27 0.000695	ETPVEDDESC Type of fence	28
0.000574	CTPLGDCDDI Type of street	29 0.000563	CESTADCODN State of
residence			
30 0.000471	NESTADNMCO State of spouse	31 0.000471	
CESTADUFCO State of origin of spouse	32 0.000453	EOCUMODESC Occupation	
of housing	33 0.000371	CREQUEFECH Type of occupation of the	
applicant's housing			

Order	Score	Variable	Description
0.000329	ETPPI	SDESC Floor Type	
35	0.000304	ETPCOBDESC	Coverage type
36	0.000303	EGLETEDESC	Plot of land
37	0.000197	FREQUESINE	SINE registration indicator
38	0.000158	ETPCONDESC	Housing construction status
39	0.000143	CREQUEAGUA	Tap water indicator
40	0.000126	CREQUEENER	Electricity indicator
41	0.000119	FREQUESINC	Spouse's SINE Registration Indicator
42	0.000103	CREQUESANI	Basic sanitation indicator
43	0	VREQUEPERC	Value of per capita family income
44	0	AREQUENUCO	Number of rooms
45	0	QREQUEQTDP	Number of dependents of the applicant

Table 27 - Result of Information Gain Attribute Ranking for variables.

- Pearson coefficient**

Pearson's correlation coefficient measures the degree of correlation between two variables and the direction of this correlation, which may be positive or negative. Assume values between -1 and 1. Where r is the Pearson coefficient, when $r = 1$ it means that there is a perfect positive correlation between the two variables; when $r = -1$ it means that there is a perfect negative correlation, which means that when a variable increases the other decreases in the same proportion. On the other hand, the closer the r is zero, the less linearly dependent the two variables are [HaK06].

The calculation of Pearson's correlation coefficient is performed as follows:
presented:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \cdot \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}},$$

Being $x_1, x_2, \dots, x_n$ and $y_1, y_2, \dots, y_n$ measured values of both variables and

$$\bar{x} = \frac{1}{n} \cdot \sum_{i=1}^n x_i \quad \text{and} \quad \bar{y} = \frac{1}{n} \cdot \sum_{i=1}^n y_i$$

being the arithmetic means of both variables.

Analysis using the Pearson correlation method was performed for the variables numerical, and some of the variables were binarized (crequesexo - sex of the applicant, frequency indicator of registration in SINE, crequeener indicator of electrical energy, crequeagua-piped water indicator, crequesani indicator of basic sanitation) or discrete (egrauidesc applicant's level of education, egrauidsc spouse's level of education) before being submitted for analysis. The Appendix E presents the results found for the sixteen variables submitted to the method. Table 28 shows the largest correlations found.

Variable 1	Variable 2	Correlation
QREQUEQTDP (quantity of dependents of the applicant)	AREQUEMEMB (number of members of the family)	,785(**)
VREQUEPERC (value of <i>per capita</i> income of the family)	VREQUEREND (family income value)	,774(**)
QREQUEQTDP (amount of dependents of the applicant)	QREQUE0A15 (amount of dependents up to 15 years old)	,592(**)

Table 28 - Variables with highest levels of correlation.

- Choice of variables**

The use of the ROC curve to select the most relevant variables, demonstrated that variables such as CREQUEIDAD (age of the applicant) and

NREQUEIDDC (age of applicant's spouse) are good classifiers for the target variable TYPE.

The use of Information Gain to select the most relevant variables, demonstrated the predictive capacity of each of the forty-five variables considering the TYPE variable as the target variable.

Using Pearson's Correlation Coefficient to select variables most relevant, demonstrated the degree of correlation between each of the sixteen numerical variables.

Based on the three types of analyses performed, the lowest relevance was observed of the variable QREQUEQTDP (number of dependents, regardless of the receipt or not of the benefit, of the applicant) to classify the problem. It is if a variable created with the objective of optimizing the modeling process through of the RNA, however the correlation analysis of the data, using the Pearson method demonstrated the high degree of correlation (0.785) of this variable with the variable AREQUEMEMB (number of family members), where QREQUEQTDP by the information gain method, it had one of the worst classifications, thus it was decided to remove it. The choice to remove just one variable is made by choosing to let the ANN, with its robustness and generalization power, handle appropriately the remaining variables in the experimented model.

5.4.4 Descriptive Statistics of Data

Once the attributes (variables) to be modeled were defined, a exploratory data collection through statistical rules for both categories of variables: numerical and categorical.

- **Numeric Variables (continuous)**

Minimum, maximum, average, standard deviation and deviation values were collected. standard mean for the 15 numerical variables: AREQUEMEMB, AREQUENUCO,

CREQUEAGUA, CREQUEENER, CREQUEIDAD, CREQUESANI, CREQUESEXO,
 EGRAUIDESC, EGRAUIDSCC, FREQUESINE, NREQUEIDDC, QREQUE0A15,
 QREQUEAP16, VREQUEPERC and VREQUEREND.

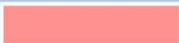
Attribute	Description	Value Minimum	Value Maximum	Average	Detour- Standard
AREQUEMEMB	Number of family members	2	20	5.3387	2.019
AREQUENUCO	Number of rooms	1	7	3.5326	1,642
CRACKWATER	Tap water indicator	0	1	0.836	0.3703
CRACKER	Electricity indicator	0	1	0.951	0.2158
CREQUEIDAD	Applicant's age	15	93	41.9088	9.0253
CREQUESANI	Basic sanitation indicator	0	1	0.7656	0.4237
CREQUESEXO	Applicant's sex	0	1	0.0839	0.2773
EGRAUIDESC	Applicant 's level of education	0	9	3.0375	1,356
EGRAUIDSCC	Applicant's spouse 's level of education	0	8	1.2542	1.7025
FREQUESINE	SINE registration indicator	0	1	0.0424	0.2014
NREQUEIDDC	Age of the applicant's spouse	5	99	44.5337	6.4377
QREQUE0A15	Number of dependents up to 15 years old.	0	11	1.8319	1.4934
QREQUEAP16	Number of dependents aged 16 and over.	0	13	2.1212	1.6268
VREQUEPERC	Family per capita income value	0	302	43.4878	27.2032
VREQUEREND	Family income value	0	1580	217.51	137.292

Table 29 - Data exploration of numerical variables.

- Categorical (discrete) variables**

Gini values, quantity and percentages of each class were collected.

30 categorical variables: CREQUEORGA, CREQUESCON, CESTADUFCO, CREQUEORCO, CESTADUFIC, FREQUESINC, CESTADCODN, CREQUEFECH, FREQUEBFAM, CTPLGDCODI, EBAIRRNOME, EPROPFDESC, NESTADNOMC, NESCIVNAME, ECBOTBDESC, ECBOTBDSCO, ESITTRDESC, ESITESDESC, NESTADNMC, NESTADNMIC, ECBOTBDSCC, EOCUMODESC, ETPVEDDESC, ETPPISDESC, ETPCOBDESC, EGLETEDESC, ETPCONDESC, EORIGEDESC, EBAIRRNESC and type. Some results are shown in Table 30. The result of the statistical analysis of some of the thirty categorical variables is in Appendix F.

Results				
Attribute	Gini	Distribution		
ETPPISDESC (Floor type)	0.6506	Values	Count Percent	Histogram
		SUBFLOOR	3466 32.21 %	
		BRICK/CEMENT 5072	47.13 %	
		TERRABATIDA	1498 13.92 %	
		CERAMICS	700 6.50 %	
		STILTS	26 0.24 %	
ETPCOBDESC (Type of coverage)	0.6818	Values	Count Percent	Histogram
		SLAB	2183 20.28 %	
		CERAMICS TILES 5085	47.25 %	
		ZINC/BRASILIT 2070	19.23 %	
		PLASTIC/CARNABY 1396	12.97 %	
		IT DOES NOT HAVE	28 0.26 %	
EGLEETEDESC (Type of land of the earth)	0.7074	Values	Count Percent	Histogram
		TENANT 1567 14.56%	portion of land)	

		OWNER	4725	43.90 %	
		PROPPARENT	1831	17.01 %	
		TENANT	2385	22.16 %	
		PROPPATRON	254	2.36 %	
ETPCONDESC (Situation of 0.7614 housing)		Values	Count	Percent	Histogram
		FINISHING	2795	25.97 %	
		READY	3264	30.33 %	
		PARALYZED	2815	26.16 %	
		SEMINFORMATION	827	7.68 %	
		STARTED	512	4.76 %	
		OTHERS	549	5.10 %	

Table 30 - Data exploration of categorical variables.

For every set of applicants classified as successful or unsuccessful there is a predominance of their housing characteristics. Most of the buildings have cement floors and ceramic tile roofs, and the applicants or spouses the owners of the portion of land and the housing is with the construction completed.

5.4.5 Discretizations and Normalizations

Then, the categorical fields were encoded into numeric fields. to match the neural network input.

• Data Discretization

Through the analysis of the histograms, intervals were defined in order to facilitate modeling by ANN. According to Kamber [HaK06], some techniques for discretization of data can be used, among them equality of frequency. In an equality of frequency in histogram, bars they are created so that the frequency of each segment is constant.

Attributes such as CREQUEORGA (Issuing body of the identity card of applicant), CTPLGDCODI (street type code), NESTADNOMC (State of origin of the spouse) and ESITESDESC (code for the special situation of dependents) because they are categorical in nature, they were discretized taking into account the occurrence (*count*) of each class as shown in Appendix G.

- **Data Normalization**

All numeric attributes in the range [0,1] were normalized by linear interpolation which consists of considering the minimum and maximum values of each attribute in the scaling adjustment.

According to Kamber [HaK06], the most commonly used normalization techniques are:

- Linear Normalization: in this technique, the distribution of the values of attribute follows a linear function, preserving the relationship between the data originals.

$$x_{normalized} = (x - x_{min}) / (x_{max} - x_{min})$$

where x is the normalized value corresponding to the initial value x of attribute; x_{min} is the minimum value among all attribute values and x_{max} is the maximum value among all attribute values.

- Normalization by Sum: In this technique, the value of the variable is divided by the sum of the attribute values.

$$X_{normalized} = x / \sum x$$

- Maximum Normalization: In this technique, the highest value of the attribute is taken for the normalization factor.

$$X_{normalized} = x / x_{max}$$

- Normalization by Score: In this technique, the central axis is shifted to the average of the attribute values and normalizes according to the standard deviation.

$$x_{normalized} = (x - \bar{x}) / \sigma$$

- **Converting Categorical Attributes to Numeric Attributes**

The categorical attributes EGRAUIDESC and EGRAUIDSCC were transformed in numeric to match the ANN input:

Classes	Value
Illiterate	1
Up to the incomplete 4th grade of elementary school	2
With completed 4th grade of elementary school	3
With completed elementary education	4
From 5th to 8th grade of incomplete elementary school	5
Completed high school	6
Incomplete high school education	7
NOT INFORMED	0
Completed higher education	8
incomplete higher education	9
N/A	0

Table 31 - Values for each class of the variables EGRAUIDESC and EGRAUIDSCC.

The N values of the attribute domain are represented with N bits. each category is discrete at a value from 1 to N and is represented by a string of digits binaries.

For this task, the binary representation technique by temperature was used. [AVL99]. The temperature code is most often used when the values discrete variables are related in some way. If, for example, a variable of a discrete nature discrete can assume values (illiterate, up to the incomplete 4th grade of education fundamental, incomplete higher education, complete higher education); it is desired that the difference between illiterate and up to the incomplete 4th grade of elementary school is lower, while the difference between illiterate and having completed higher education is as large as possible. The code thermometer for illiterate will be represented as 1000, while higher complete will be 1111 and incomplete higher 1110. The similarity of the domain values is verified using the *Hamming Distance*, which counts for each position *i* of the bit strings if they are equal or different. If different, one is added to one

counter. In the observed example, the *Hamming* distance between the values illiterate and full upper is equal to three since the second, third and fourth bits are different.

The categorical fields CREQUEAGUA, CREQUEENER, CREQUESANI and FREQUESINE were converted into binary numbers, by direct relation, each being category converted to an attribute that can assume value 1, if the instance has the corresponding category , or 0 otherwise.

6 PBEM Data Mining Modeling and Evaluation

This chapter covers the modeling and performance evaluation phases. according to the CRISP-DM methodology , in addition to the cost of errors and successes in the model.

6.1 Modeling

To carry out the assessment of the criteria for granting credit to the program Bolsa Escola, the Multilayer Perceptron (MLP) neural network model was used, trained with the backpropagation algorithm [Hay99], [DGR86], model used with success in applications of pattern classification problems. Among the most attractive features of this type of neural network is the excellent generalization capacity, the simplicity of network operation and the fact that it is a universal function approximator [KMH89].

Before the start of the ANN training, it was developed in language C programming, a program to perform data separation, dividing randomly split the entire data set into ten disjoint subsets (folds), respecting the same proportion of data for the classes of the target variable Type (90% Success and 10% Failure) [HaK06]. Each of the ten subsets created was used as a test set (these being an independent sample of the data used in the model construction) and the other nine subsets were assembled into sets of training and validation. The process was repeated ten times, where each of the 10 generated sets was used to evaluate the performance of the ANN. This procedure is known as *K-fold CrossValidation* . Stratified).

In a subsequent stage, the possibility of providing greater stability to the system by the Monte Carlo method using a number of iterations

that, for the work, five were considered. Thus, the procedure described above was redone four more times to obtain five resulting iterations. The the result of all iterations generated a total of fifty samples for training, evaluation and testing to be submitted to the RNA. In this way, five evaluations for each instance of the available database. Subsequently, the median of the scores and with this the final performance of the system was obtained .

6.1.1 N-Fold Cross-Validation

This method divides the data set into n subsets , one of which subset is separated and used as the test set, while all others are used in the ANN training process. This process is repeated n times with each of the n subsets being used as a test set only once [KOH95]. An advantage of this method is that all examples in the data set are used for training and testing (Illustration 12).

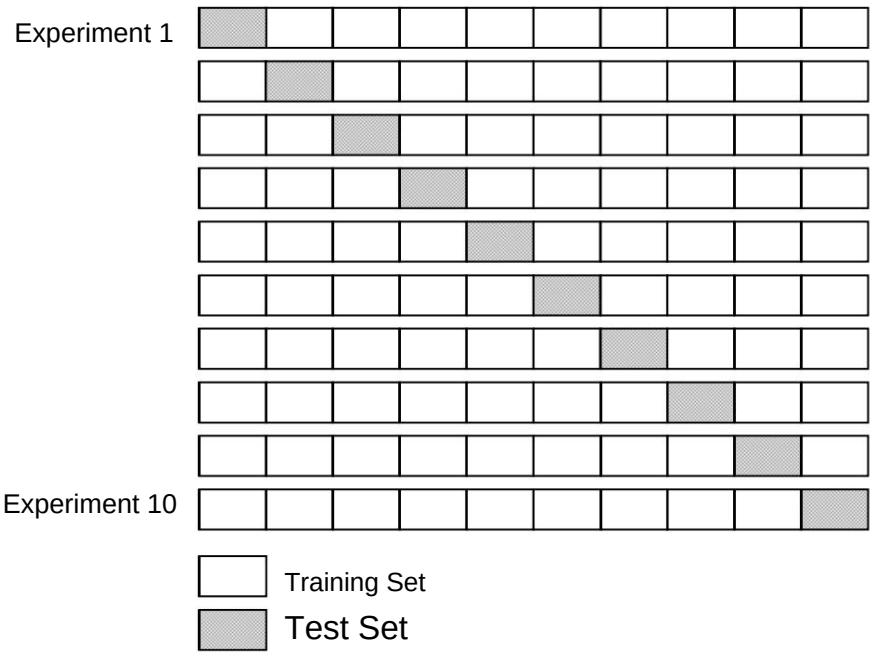


Figure 12 - Cross-Validation method with 10 folds

6.1.2 Monte Carlo

Monte Carlo is the procedure that investigates the random distributions of various statistics. It is a branch of mathematics that is concerned with experiments with numbers random [H&H64]. This method is mainly used in two applications, one is to evaluate the empirical random distribution of a statistic and another is to study the effects of violating assumptions that are hidden in some statistics.

The main objective is, through simulations, to create an environment in which information about possible alternative actions can be obtained through experimentation, quickly and at a low cost.

For this work the Monte Carlo method is used to solve a problem²⁰ through a series of random simulations. The accuracy of the result is final dependent on the number of attempts. Such a balance between the accuracy of the result and the computationally feasible time led to the choice of the number of five iterations for application of simulations.

6.1.3 Network Training and Topology

In the final stage of modeling is the construction of the ANN model to be used for the binary classification problem. Problems were taken into account that may arise in training, with inappropriate choices of the parameters used in the model (Table 32).

Various combinations in learning rate, momentum, stopping criterion, number of layers and neurons in the intermediate layers were experimented with. For the final choice of such combinations, the percentage of correct answers was considered. target class and training time performed on one of the fifty subsets of data.

²⁰ Problem: Provide greater stability to the system.

Settings	Greatest extreme	Minor Extreme
Rate of Apprenticeship	Instability around the best solution	Excessive delay in training
Criteria of Stop	Reduces the risk of local minimum, speeds up training and increases the risk of instability	No local minimum risk suppression effect .
No. of cycles	Poor generalization and excessive delay in training	Does not reach knowledge

Table 32 - Risks of extreme values for ANN training.

The neural network used in the experiment of this work was the MLP network with a single hidden layer, since according to Cybenko [Cyb89], these networks can generalize any linearly continuous function. From primary experiments, as discussed in the previous paragraph, we chose 5 neurons in the layer hidden, using the WEKA tool²¹. As a learning algorithm it was Backpropagation was used, with a learning rate equal to 0.001, momentum 0.1, and stopping criterion based on the number of epochs: 10,000 and the increment of the *Mean Squared error* (mean squared error) referring to the validation set as a metric.

To carry out the fifty experiments resulting from the *ten-fold cross-validation* and the Monte Carlo method, four computers were needed (each one with distinct processing capacity) running the training of the twenty-one ANN and four hours a day, for about twelve days.

Considering that the problem was modeled as categorical and in 2 classes, the response of each neuron can be conjugated by means of a linear transformation into a single scalar quantity (the score) that is defined in the continuous domain between 0 and 1, with zero representing Success and unity (100%) meaning "Failure. This representation of the response by a scalar quantity allows a much more refined monitoring of network performance.

²¹ WEKA. Open source data mining tool. <<http://www.cs.waikato.ac.nz/ml/weka>> Accessed in November/2008.

6.1.4 Mean Square Error

The mean squared error, in the field of statistics, of an estimator is one of the ways of quantifying the amount by which an estimator differs from the true value of the quantity to be estimated. The difference occurs due to randomness or simply because the estimator does not take into account information that could generate a more accurate estimate [MFD74].

The error of the ANN output layer is given $E_p = 1/2 \sum_{j=1}^n (t_j - O_{pj})^2$, where O_{pj} is the response to the neuron and n_j and t_{pj} is the desired output.

6.2 Evaluation

Transforming the output of the multilayer perceptron network into a response continuous for the binary decision problem (score) demands a threshold (point of cut) to define the separation of classes [SBRN04]. The cut point used was obtained by KS2Max. The metrics used here to measure performance are the graph of the Kolmogorov-Smirnov (KS2) statistical test [Con99], the ROC curve [ProF98], the GINI coefficient [Hoff80] and the confusion matrix [Kan03].

6.2.1 Kolmogorov–Smirnov

Also known as KS, the *Kolmogorov-Smirnov* test [Con99] compares, through cumulative distribution function, data from two distributions indicating, in this work, the successes and failures.

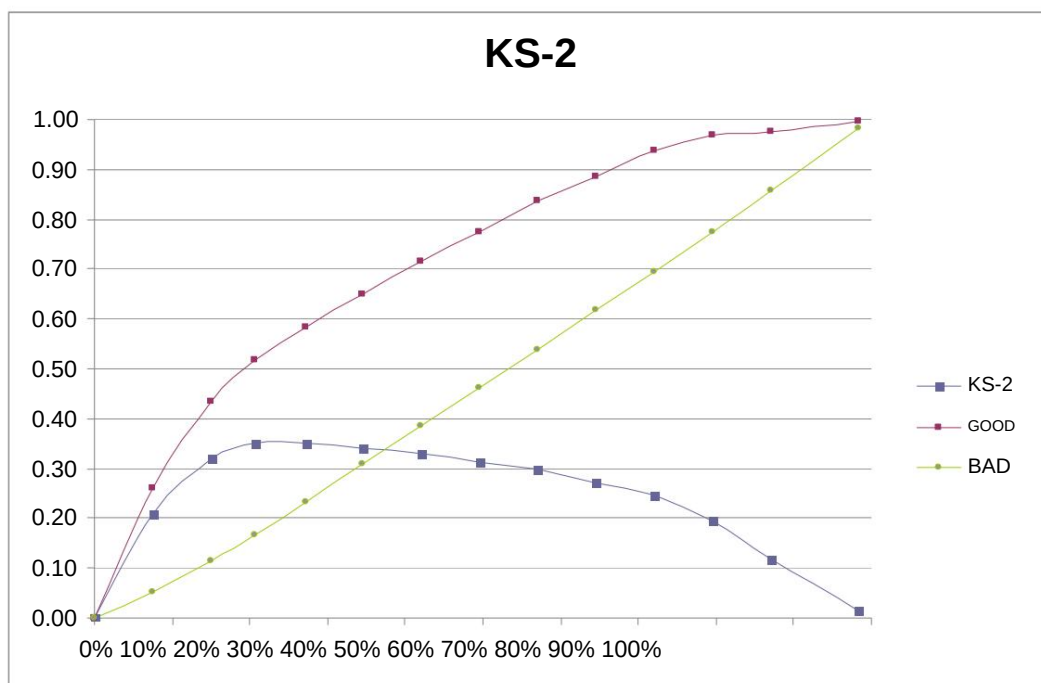
In binary classification problems the KS2 curve is the difference between two cumulative distribution functions with the score as the independent variable. One distribution contains the score of the positive class and the other of the negative class. It has been used to evaluate the performance of binary classifiers indicating the difference between the distributions of bad and good examples, that is, it assesses how well

the model can distinguish examples classified as good from examples classified as bad [Con99].

From the KS it is possible to evaluate the performance of the classifier from different cutoff points, so as to make the final decision of the classifier be taken more smoothly and optimally. The optimal cut-off point is the point of maximum distance between the two classes, which means ensuring the highest percentage of hit.

Therefore, for this work, the cutoff point used was KS2Max. The cutoff point KS2Max cutoff was also used for the confusion matrix presented in section 6.2.4

From Graph 4, it can be seen that there is a point of greater separation between the distributions of the scores of the occurrences of success and failure. This point indicates that the system is able to predict failure with a good margin of accuracy. The measurement of KS2Max for the test set was 0.3536 and the area (AUC) was 0.2470.



Graph 4 - KS graph for RNA.

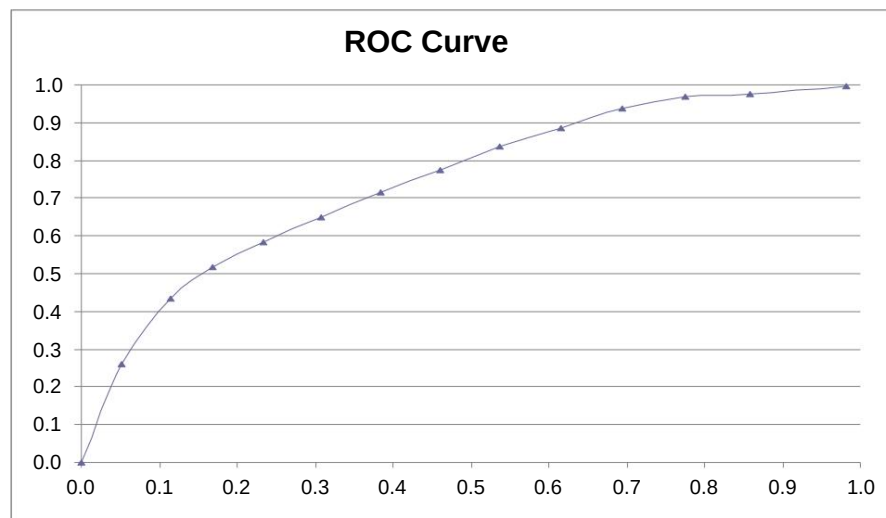
6.2.2 ROC Chart

The ROC (*Receiver Operating Characteristics*) graph is used to evaluation and selection of the most significant variables, was also used to measure of the performance of the experimented model.

ROC curves show the relationship between Error I and sensitivity (concepts these will be seen in section 6.2.4) which corresponds to the relationship of false rates positives (FP) and true positives (VP) [ProF98]. This relationship predicts the behavior of classifiers regardless of costs and distribution of classes. For each cutoff point, the sensitivity and specificity complement (1 – specificity) (these concepts will also be seen in section 6.2.4) are calculated and placed on each respective axis of a two-dimensional graph. The Graph 5 illustrates the ROC curve of the work.

The result obtained for this work was an area of 0.7470, a result that represents a higher probability of success for classifying random examples of true positives [FAW05].

For work, the result concerns the greater probability of success as to the classification of instances considered unsuccessful in the PBEM.

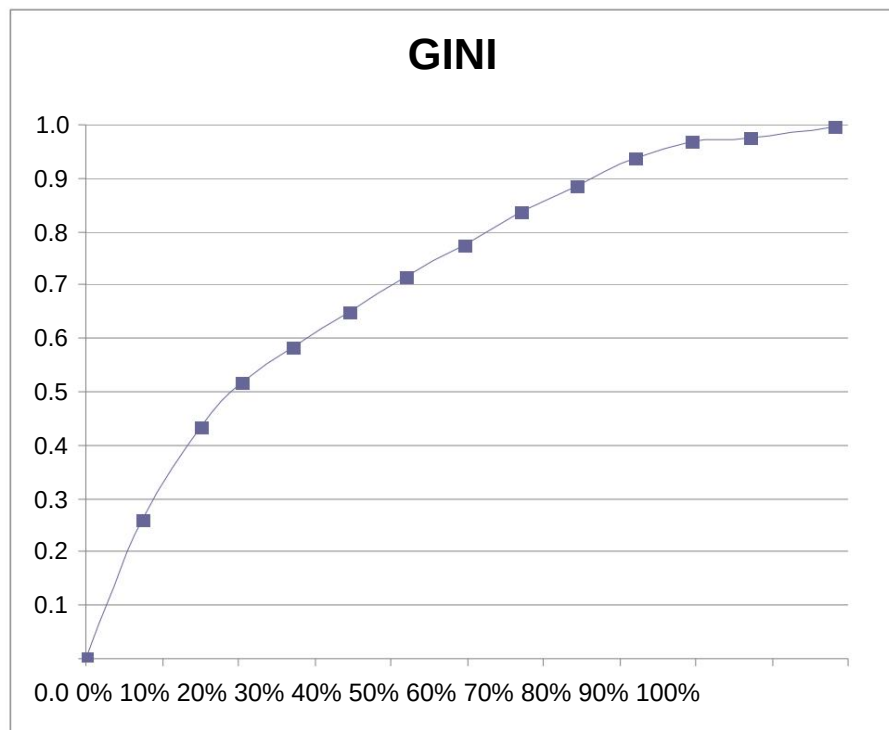


Graph 5 - ROC Curve Graph.

6.2.3 GINI coefficient

The Gini coefficient is a measure of inequality developed by the Italian statistician Corrado Gini. It is based on the *Lorenz curve* and is commonly used to calculate inequality of income distribution, but can be used for any distribution. Numerically, it varies between 0 and 1, where 0 corresponds to complete income equality (where everyone has the same income) and 1 corresponds to complete inequality (where one person has all the income, and the others have nothing) [Hoff80]. The result obtained for this work was an area of 0.4405 (Graph 6.3), with the x-axis corresponding to the population and oy corresponding to the target class (failure).

For the work, the result demonstrates a significant quality of classifier to classify instances as success or failure.



Graph 6 - Lorenz Curve Graph.

6.2.4 Confusion Matrix

A confusion matrix compares the desired real cases with the classification carried out by the [Kan03] system and for this reason, it is used to evaluate the *performance* of the classifiers. The result of the classification of a given instance results in one of four possible situations. Considering the hit as *positive* and the error as *negative*, if an instance is *positive* and is classified by the system as such, we have a *true positive situation*; if an instance is *positive* and is classified by the system as *negative*, there is a false negative situation . Similarly, if an instance is *negative* and is classified by the system as *negative*, there is a *true negative* situation and if an instance is *negative* and is classified as *positive* we have the situation of *false positive*. Taking these four as a basis possible situations for each instance, a confusion matrix can be assembled (Table 33) which will be used to extract metrics such as specificity, sensitivity and accuracy.

Actual / Expected	<i>Positive</i>	<i>Negative</i>
<i>Positive</i>	<i>True Positive</i>	<i>False Positive</i>
<i>Negative</i>	<i>False Negative</i>	<i>True Negative</i>

Table 33 - Confusion Matrix

TP (*True Positive*) Number of correct predictions for the cases that are actually positive.

FN (*False Negative*) Number of wrong predictions for the actually true cases positive.

TN (*True Negative*) Number of correct predictions for the cases that are actually negatives.

FP (*False Positive*) Number of wrong predictions for the actually positive cases negatives.

For this work:

TPR (True Positive Rate). Number of instances correctly classified as a failure in relation to all the failures of the base.

FNR (False Negative Rate). Number of instances misclassified as a success in relation to all the failures of the base, complementary to the TPR.

TNR (True Negative Rate). Number of instances correctly classified as a success in relation to all the successes of the base.

FPR (False Positive Rate). Number of instances misclassified as a failure in relation to all the successes of the base, complementary to TNR.

From the confusion matrix, two error measures can be obtained:

- Error I: corresponds to the percentage of applicants who are successful cases and are classified as unsuccessful. It is given by the formula $FP/(FP+TN)$. There is also another measure that is equivalent to this one.

Specificity measures the classifier's hit rate on the applicants that represent success. It is given by the formula $TN/(FP+TN)$.

- Error II: corresponds to the percentage of applicants who are unsuccessful cases and are classified as successful. It is given by the formula $FN/(FN+TP)$. There is also another measure that is equivalent to this one.

Sensitivity measures the classifier's hit rate on the

applicants representing failure. It is given by the formula $TP/(FN+TP)$.

The total hit rate of the classifier is given by the accuracy measure (*accuracy*) through the formula $(TP+TN)/(TP+FN+TN+FP)$.

The confusion matrix for this work is represented in Table 34.

	Failure	Success
Failure	701	2.173
Success	463	7,425

Table 34 - Confusion Matrix for work

The respective error and hit rates of the confusion matrix are presented in Table 35.

	Failure	Success
Failure	TPR=0.60	FPR=0.23
Success	FNR=0.40	TNR=0.77

Table 35 - Error and hit rates of the confusion matrix

The measures, for the work, extracted from the confusion matrix are shown in Table 36.

Error I = 0.23	Error II = 0.4
Specificity = 0.77	Sensitivity = 0.6
Specificity = $1 - 0.23 = 0.77$	Sensitivity = $1 - \text{Error II} = 1 - 0.4 = 0.6$
Accuracy = 0.685 or 68.5%	

Table 36 - Other performance measures

The total hit rate (68.5%) demonstrates the good performance of the classifier. for the PBEM.

6.2.5 Cost

The cost analysis was interpreted taking into account that the average number of dependents per applicant family is 2.7, that is, more than one dependent per applicant family. Since the benefit amount is one minimum wage (MW) in force for families with more than one child in school, we consider this value for the

analysis. On the other hand, the representative social cost was also considered ²² (CSR) as the value to be considered in the event of errors in the classification operation. The Table 37 lists the costs involved in classification.

Classification	Associated Cost
TP - Classify correctly as "Failure"	0.00/Req.
FP - Misclassify as "Failure"	CSR
TN - Classify correctly as "Success"	1 SM/Req.
FN - Misclassify as "Success"	1 SM/Req. + CSR

SM: Current Minimum Wage
CSR: Representative Social Cost

Table 37 - Costs involved in RNA classification.

Another way to analyze the costs involved is by using Game Theory.

Game Theory is the science of decision making in situations of interdependence.

[Fian06]. Unlike a unilateral decision, here it is a question of deciding taking into account account of decisions of other(s) involved in the same decision problem. The costs were not associated because there was no value for the social cost representative defined in the work.

Therefore, the results matrix, also called the payoff matrix, has representation as shown in Table 38.

		System	
		T	F
Real	P	TP(0.00/Req.)	FP(-CSR)
	N	TN(1 SM/Req.)	FN(-1 SM/Req.- CSR)

Table 38 - Matrix with results of the costs involved.

²² Representative Social Cost: concept suggested as a way of measuring the possible social costs involved in not benefiting a dependent who is a success story in the PBEM or benefiting a dependent which is a case of failure. It would be, for example, the cost to society of having a young person involved in crime.

7 Final Considerations

The motivation for this dissertation came from the availability of data on the Bolsa Escola Program of the City of Recife, as well as the possibility to apply data mining processes in order to generate social benefits significant.

7.1 Conclusions

The process of granting the benefit once supported by the proposed model, could generate a reduction in losses to the public coffers, since the decision to granting to an applicant will be anticipated by the model used to classify the instances as being Success or Failure. ” “ ”

The Directorate of Social Support for Education of the City of Recife in together with EMPREL²³, provided the necessary database and clarifications to start and continue the work.

All stages of the project were addressed using the CRISP-DM methodology data mining with the exception of the distribution step which is not part of the scope of this work.

The understanding of the business was widely discussed with the team. Directorate of Social Support for Education of the City of Recife, which was always available to resolve any doubts that might arise.

Understanding the data composed of a database with 184 tables required a considerable amount of time, since the employees responsible for the data were involved in other projects, making it impossible to clarify doubts in a fast. The main source of extraction of variables was the tbRequerente table with all the

²³ Municipal IT Company of Recife.

registration data of the applicant for the benefit of the Municipal School Grant Program;
This table was made from a statistical survey of the completion of various fields.

In the data preparation phase, the entities tbDependent and tbEvolucaoFamilia were also used to find out the number of dependents per applicants receiving the benefit and the reason for the absence, respectively. In data preparation, outliers and missing data were treated; they were also created new variables and through three different techniques, evaluated for selection to the process of modeling.

For the modeling stage, artificial intelligence technique was chosen using Artificial Neural Networks to model the problem. This step was also exhausting, taking several days, especially since the option was chosen for Monte Carlo method with the aim of providing greater stability to the system; several tests were carried out varying combinations of Artificial Neural Network configurations before defining the final ANN model to be applied to the fifty subsets.

The last stage of CRISP-DM at work, the performance evaluation, was shown through the Komogorov-Smirnov test, ROC and Lorenz curves, in addition to model to measure the costs of correct and incorrect choices of the classifier using the Game Theory. The results of the evaluations were satisfactory and demonstrated the good performance of the classifiers to predict the Failures of the base.

7.2 Validation

The objective of the work was achieved by demonstrating the good performance of the classifiers using data mining techniques, with artificial neural networks, and by the acceptance by the specialist in the application field. The result comes confirm some empirical findings²⁴ as the socio-economic influence on success and failure cases in PBEM.

²⁴ Findings based on lived experiences.

7.3 Limitations

There were limitations in the model constructed, mainly due to the absence of data updated in several tables, which made it impossible to use other variables that could represent a gain for the artificial neural network. Another limitation is the lack of a data dictionary for the various entities that make up the database, in addition to a lack of documentation of the SEBE system for initial understanding of the data which may have resulted in the non-use of other information relevant to the problem.

7.4 Future Work

The results obtained enable the carrying out of other works complementary futures based on the Bolsa Escola Program of the Municipal Government from Recife as:

- The development of software that implements the constructed model , way to support the evaluation committee in real time in the decision on the granting or not granting the benefit;
- Correlation with databases from other Secretariats and Agencies such as way to better assess the reasons for absences (health, safety public and others) and create variables that can improve performance of RNA;
- Use of other data mining techniques with the aim of evaluate and find better hit levels for the Failure target class.
- Study to survey the representative social cost proposed in this work.

8 Bibliographic References

[AR99] AMARAL, C; RAMOS, S. Minimum Income and School Grant Programs : Overview and Perspectives. Current and (1999). Available at in: <
http://www.iets.org.br/biblioteca/Programas_de_renda_minima_e_Bolsa_Escola.pdf> Accessed on: 10 Nov 2008

[CS03] Cardoso, E.; Souza, AP The impact of income transfers on child labor and school attendance Brazil. in They are Paulo: 2003. <http://www.econ.fea.usp.br/elianacardoso/ECONBRAS/cardoso-souza.pdf>. Accessed on September/2008.

[PNAD07] Home Search. National put Sample of
<http://www.ibge.gov.br/home/estatistica/populacao/trabalhoerendimento/pnad2007/comentarios2007.pdf>. Accessed on March 20, 2009.

[MS99] MAMBRINI, J.; CÉSAR, CC; SOARES, JF Determining Factors of Performance of Students from Minas Gerais in the 1995 SAEB. In: 1st Latin American Conference on Applied Mathematics Statistic Annals, Carlos August. 1999..<<http://www.fde.ufmg.br/game/saeb95.pdf>>. Accessed in November/2008.

[Const98] Constitution of the Federative Republic of Brazil of 1998. Website:
http://www.planalto.gov.br/ccivil_03/constituicao/constitui%C3%A7ao.htm. Accessed on March 10, 2009.

[ECA09] Child Site Status from the and of Adolescent.
http://www.planalto.gov.br/ccivil_03/leis/l8069.htm. Accessed on March 10, 2009.

[TWB44] World Bank website. World<http://www.worldbank.org/>. Accessed on March 10, 2009.

[UNDP06] United Nations Development Programme
http://www.pnud.org.br/pobreza_desigualdade/reportagens/index.php?id01=2388&lay=pde. Accessed on March 10, 2009.

[Cos98] COSTA, Alfredo B. (1998). Social Exclusions. Lisbon: Gradiva Edition (Series Democratic Notebooks).

[SEPLAG08] Secretariat of Planning and Management of the State of Santa Catarina.
http://www.seplag.rs.gov.br/principal.asp?conteudo=indicadores&act=view&cod_menu=132&cod_indicador=13&cod_menu_esq=123. Accessed on March 10, 2009.

[HDI08] Human Development Indexes.
(2008).http://hdr.undp.org/en/media/HDI_2008_EN_Tables.pdf Accessed on 27 March 2009.

[HDR08] Human Development Reports
http://hdrstats.undp.org/2008/countries/country_fact_sheets/cty_fs_BRA.html.
Accessed on March 27, 2008.

[PCR08] Place from the Recife City Hall. <http://www.recife.pe.gov.br/2008/08/21/bolsa-escola-recebe-novos-beneficiarios-163535.php>. Access in December/2008.

[FSS96] Fayyad, U., G. Piatetsky-Shapiro, and P. Smyth, (1996). *The KDD process for extracting useful knowledge from volumes of data*. Common. ACM,

[FPS96] Fayyad, U., Piatetsky-Shapiro, G., & Smyth, P. (1996). Knowledge Discovery and Data Mining: Towards an Unifying Framework. *In Proceeding of The Second International Conference on Knowledge Discovery and Data Mining (KDD'96)*.

[AVL99] Aurélio M.; Vellasco M.; Lopes, Carlos H..(1999) Discovery of Knowledge and Data Mining . IAC Laboratory of Applied Computational Intelligence. PUC-Rio.

[McP43] W. S. McCulloch and W. Pitts. (1943). A logical calculation of the ideas immanent in nervous activity. *Bulletin of Mathematical Biophysics*.

[Hay99] Haykin, Simon..(1999). *Neural Networks: A Comprehensive Foundation*. 2nd Edition. Prentice Hall International, Inc.. New Jersey, USA.

[BCL07] Braga, Anthony;
Artificial Neural Networks: Theory and Applications. Rio de Janeiro: LTC, 2007.

[Cor02] Cortez P., "Nature-Inspired Models for Series Forecasting Temporals", Doctoral Thesis, University of Minho, 2002.

[MM70] JM Mendel and RW McLaren. (1970). *Adaptive, learning, and pattern recognition systems; theory and applications*, chapter Reinforcement-learning control and pattern recognition systems. NW.

[Ros58] F.Rosenblatt. (1958). The perceptron: A probabilistic model for information storage and organization in the brain. *Psychol*.

[MiP69] M. Minsky and S. Papert. (1969). *Perceptrons: An introduction to computational geometry*. MIT Press, Massachusetts.

[Cyb89] G. Cybenko.(1989). Approximation by Superpositions of a sigmoid function. *Mathematics of Control, Signals and Systems*.

[Cru07] Cruz A.(2007). Data Mining via Artificial Neural Networks and Machine Learning Support Vectors. Master's Dissertation. University of Minho.

[Fah98] Fahlman S. (1998). Fast Learning Variations on Back-Propagation: An Empirical Study, In D. Touretzky GH and Sejnowski, T., editors, Proceedings of Connectionist Models Summer School. Los Altos CA, USA. Morgan Kaufmann Publishers.

[ZuG93] Zupan J. and Gasteiger J.(1993). Neural Networks For Chemists: An Introduction , VCH, New York.

[WiH00] Wirth, R. and J. Hipp. (2000). CRISP-DM: Towards a standard process model for data mining,in In Proceedings of the Fourth International Conference on the Partial Application of Knowledge Discovery and Data Mining (PADD00).

[Cun05] Cunha R. (2005). Methodology for Developing Solutions in Data Mining: A Practical Study in Fault Diagnosis. Dissertation of Master's degree. UFPE.

[Cri96] CRISP-DM. (1996). Cross Industry Standard Process for Data Mining. Available at: <http://www.crisp-dm.org/Overview/index.htm>. Accessed on: 12 Nov 2008.

[CCK00] CHAPMAN, P; CLINTON, J.; KERBER, R; KHABAZA, T.; REINARTZ, T.; SHEARER, C.; WIRTH, R. CRISP-DM 1.0 Step by Step Data Mining Guide. SPSS Inc. 2000.

[RP05] RC PRATI AND P. FLACH, ROCCER: an algorithm for rule learning based on ROC analysis, in Proceedings of the Nineteenth International Joint Conference on Artificial Intelligence (IJCAI'2005), Edinburgh (UK), 2005.

[ProF98] PROVOST F., & FAWCETT, T. Robust classification systems for imprecise environments. In Proc. 15th Nat. Conf. on Artificial Intelligence. 1998. <<http://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.36.5123>> Accessed on November/2008.

[FAW05] Fawcett, T. (2005). An introduction to roc analysis. Pattern Recognition Letters.

[Mer07] Merschmann L. (2007). Probabilistic Classification Based on Analysis of Patterns. Doctoral Thesis. Niterói, RJ.

[HaK06] HAN, J., KAMBER, M.; Data Mining: concepts and techniques. USA: Morgan Kaufmann; Second Edition, 2006

[DGR86] BY RUMELHART, GE HINTON, AND RJ WILLIAMS, "Learning internal representations by error propagation", MIT Press Computational Models of Cognition and Perception Series, vol. 1, 1986.

[KMH89] K. HORNIK, M. STINCHCOMBE, AND H. WHITE, "Multilayer feedforward networks are universal approximators", *Neural Network*, vol. 2, no. 5, 1989.

[KOH95] KOHAVI, R.; A study of cross-validation and bootstrap for accuracy estimation and model selection. *Proceedings of the Fourteenth International Joint Conference on Artificial Intelligence*: 1137 1143. (Morgan Kaufmann, San Mateo).

[HaH64] Hammersley, J. M. and Handscomb, D. C. (1964). *Monte Carlo methods*. Methuen, London.

[MFD74] Mood, A., F. Graybill, D. Boes (1974). *Introduction to the Theory of Statistics* (p. 229) (3 ed.). McGraw-Hill.

[SBRN04] ROBERTO AF SANTOS, PAULO JL ADEODATO ADRIAN L. ARNAUD, RODRIGO CLV CUNHA, GERMANO C. VASCONCELOS, DOMINGOS SMP MONTEIRO - A Data Mining Application in Maintenance of Telephone Networks. SBRN, 2004.

[Con99] CONOVER, WJ, *Practical Nonparametric Statistics*, 3rd ed., John Wiley & Sons, New York, USA, 1999.

[Hoff80] Hoffman, R. (1980). *Statistics for Economists*. Pioneira, São Paulo, Brazil.

[Kan03] Kantardzic, M. (2003). *Data Mining: Concepts, Models, Methods and Algorithms*. John Wiley & Sons.

[Fian06] FIANI, R.(2006) *Game Theory*. 2nd Edition. Rio de Janeiro: Elsevier.

Appendix A

Own Goods

Description of Goods	Score 0 0
B/W TV	
BLENDER	
RADIO	0
GAS STOVE	0
IT DOES NOT HAVE	0
IT DOES NOT HAVE	0
COLOR TV	-5
SOUND EQUIPMENT	-5
BICYCLE	-5
SEWING MACHINE	-10
REFRIGERATOR	-10
TELEPHONE	-10
VIDEO CASSETTE/DVD	-10
CART	-10
WASHING	-50
MACHINE	-80
TENT OTHER LOT	-100
COMMERCIAL ESTATE	-150
ANOTHER HOUSE	-200

Number of Dependents

Amount	Score 490
8	
7	420
6	350
5	280
4	210
3	140
2	70
1	0

Coverage Type

Coverage	Scoring
PLASTIC/CARNABY	50
ZINC/BRAZILIT	25
CERAMIC TILE	10
SLAB	-25
IT DOES NOT HAVE	-25

Construction Type

Construction	Scoring
PARALYZED	20
STARTED	15
IN FINISHING	10
NO INFORMATION	0
OTHERS	0
READY	-15

Floor Type

Floor	Scoring
STILTS	50
EARTH TURF	25
SUBFLOOR	15
BRICK/CEMENT	-5
CERAMICS	-5

Sealing Type

Sealing	Scoring
WOOD	25
WOOD	25
OTHER	0
MASONRY	-25
ADOBE	-25
UNCOATED TAIPA	-25
IT DOES NOT HAVE	-25

Type of Housing Occupancy

Housing Type	Score 100
RENTED	
FINANCED	100
RENTED	100
INVASION	50
GRANTED	0
OWN	0
IT DOES NOT HAVE	0
OTHER	0

Family Income

Income Range (R\$) 0 to 30	Score 100
31 to 60	75
61 to 90	50
91 to 120	25
121 to 150	0
151 to 180	-100

Special Situation

Special Situation	Scoring
CARRIER OF DISABLED DISABILITY TO PROVIDE SUSTAINMENT	50
DISEASE CARRIER CHRONICLE	50

COATED MUD	-25
MATERIAL USED	-25

Work Situation

Situation	Scoring
UNEMPLOYED	50
HANDYMAN	50
DOES NOT WORK	50
WAGE EARNER WITHOUT A WORKING CARD	
WORKING	30
WAGE EARNER WITH WORK CARD	
WORKING	25
RURAL WORKER	25
OTHER	25
PENSIONER/RETIREE/BPC	0
SELF-EMPLOYED RURAL	0
EMPLOYER WITH SOCIAL SECURITY	
SELF-	-10
EMPLOYED	-25
EMPLOYER WITHOUT SOCIAL SECURITY	
CHILD/	-40
TEENAGERS UNDER SPECIAL PROTECTION CHILD/	50
TEENAGERS UNDER SOCIO- EDUCATIONAL MEASURES	50
CHILDREN AGED 0 TO 6 WITH MALNUTRITION	50
CHILD/TEENAGERS AT PERSONAL/ SOCIAL RISK (INF. WORKER)	50

Appendix B

Own Goods (fridge / score=-10)



Sealing Type (Uncoated Rammed Earth / Score=-25)



Coverage Type (Brasilit / score=25)



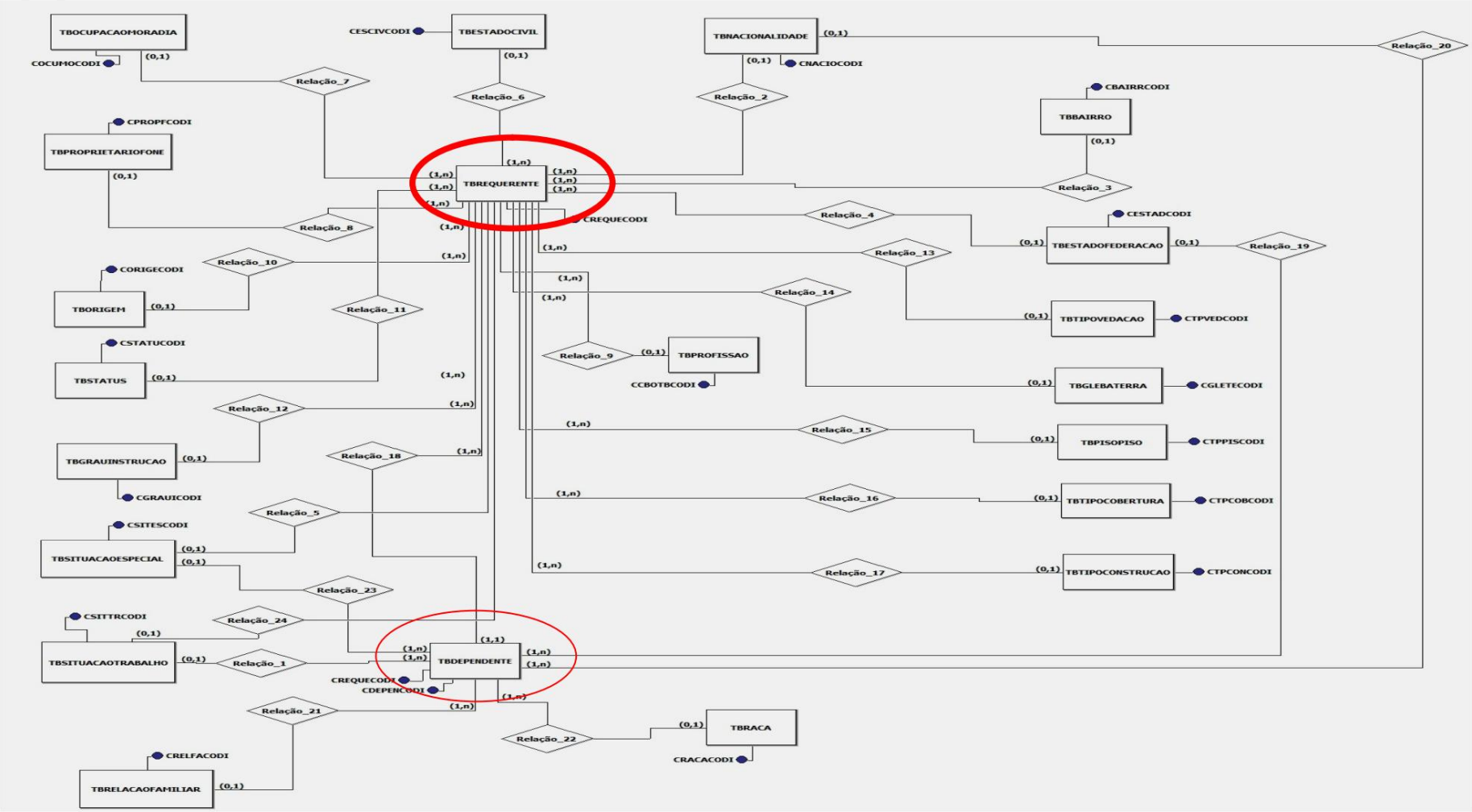
Floor Type (Dirt / Score = 25)



Sealing Type (Uncoated Rammed Earth / Score=-25)



Appendix C



Appendix D

#	FK	PK	Field Name	Field Type	Domain	Size	Scale	Subtype	Array	Not Null	Charset	Collate	Description
1			CREQUECODI	INTEGER						☒			Código do Requerente
2			CREQUEINSC	VARCHAR		10				☐	NONE	NONE	Inscrição Antiga do Requerente
3			CECOLCODI	CHAR		3				☐	NONE	NONE	Código da Escola
4			CREQUETESC	CHAR		1				☐	NONE	NONE	Tipo da Escola (E-Estadual, M-Municipal)
5			NREQUENOME	VARCHAR		40				☒	NONE	NONE	Nome do Requerente
6			CREQUESEXO	CHAR		1				☒	NONE	NONE	Sexo do Requerente
7			DREQUENASC	TIMESTAMP						☒			Data de Nascimento
8			EREQUEENDE	VARCHAR		40				☒	NONE	NONE	Logradouro
9			CREQUENUME	VARCHAR		5				☒	NONE	NONE	Número da Moradia
10			EREQUECOMP	VARCHAR		20				☐	NONE	NONE	Complemento do endereço
11			CBAIRRCODI	SMALLINT						☒			Código do Bairro
12			EREQUENCEP	CHAR		8				☒	NONE	NONE	Número do CEP
13			EREQUEREF	VARCHAR		40				☐	NONE	NONE	Ponto de Referência
14			CMUNICICEP	INTEGER						☒			Código do Município de Residência
15			CESTADCODM	CHAR		2				☒	NONE	NONE	UF de Moradia
16			CREQUEZONA	CHAR		1				☒	NONE	NONE	Zona Residencial
17			CNACIODE	SMALLINT						☒			Código da Nacionalidade
18			CMUNICICEO	INTEGER						☐			Código do Município de Origem
19			DREQUECHEG	TIMESTAMP						☒			Data de Chegada
20			CREQUEDDDF	VARCHAR		3				☐	NONE	NONE	DDD do Telefone
21			AREQUEFONE	VARCHAR		8				☐	NONE	NONE	Número do Telefone de Contato

[illegible]

Appendix E

	CREQUESEXO	FREQUESINE	CREQUEENER	CREQUEAGUA	CREQUESANI	AREQUENUCO	AREQUEMEMB	VREQUEREND	VREQUEPERC
CROCKSEX		.066(**) 1	0.01	0.015	0.011	0.006	.048(**) .039(**)	.073(**)	
FREQUESINE	1 .066(**)		0.005	0.008	.027(**)	0.018	-0.018 .044(**)	.028(**)	
CRACKER	0.01	0.005		.351(**) 1	.093(**)	.077(**)	0.015 .059(**)	.064(**)	
CRACKWATER	0.015	0.008	1 .351(**)		.242(**) 1	.153(**)	.049(**) .098(**)	.119(**)	
CREQUESANI	0.011	.027(**)	.093(**)	.242(**)		.295(**) 1	.060(**) .143(**)	.177(**)	
AREQUENUCO	0.006	0.018	.077(**)	.153(**)	.295(**)		.164(**) .148(**)	.240(**)	
AREQUEMEMB	.048 (**)	-0.018	0.015	.049(**)	.060(**)	.164(**)	1 .292(**)	.263(**) 1	
VREQUEREND	.073(**)	.028(**)	.064(**)	.119(**)	.177(**)	.240(**)	.263(**) .774(**)		
VREQUEPERC	.039(**)	.044(**)	.059(**)	.098(**)	.143(**)	.148(**)	- .292(**) 1	.774(**)	
EGRAUIDESC	0.013	.088(**)	0.008	.029(**)	.045(**)	.049(**)	- .069(**)	.056(**)	.088(**)
EGRAUIDSCC	.103(**)	-0.007	0.009	.022(*)	.059(**)	.082(**)	.138(**)	.159(**)	.057(**)
CREQUEITY	.123(**)	-.045 (**)	0.016	.035 (**)	.041(**)	.116(**)	.080(**)	.082(**)	.056(**)
QREQUEQTPD	.038(**)	-0.01	.025 (**)	.058(**)	.083(**)	.159(**)	.785(**)	.341(**)	-.146(**)
QREQUE0A15	0.006	0.004	0.007	-0.01	-0.018	-.050(**)	.456(**)	.089(**)	-.188(**)
QREQUEAP16	.040(**)	-0.016	.025(*)	.080(**)	.118(**)	.238(**)	.536(**)	.334(**)	-0.005
NREQUEIDDC	-.074(**)	-.028(**)	0.016	.032(**)	.039(**)	.112(**)	.051(**)	.027(**)	0.004

Appendix E

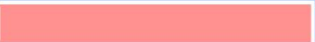
	EGRAUIDESC	EGRAUIDSCC	CREQUEIDAD	QREQUEQTPD	QREQUE0A15	QREQUEAP16	NREQUEIDDC		
CROCKSEX	0.013	.103(**)	.038(**)	-0.00712801	0.009 .025 (**)	.022(*)	0.006	1	-.074(**)
FREQUESINE	.088(**)	.058(**)	.059 (**)	.083451	.082(**)	.159(**)	.138(**)	0.004	-.028(**)
CRACKER	0.008	.785(**)	.159(**)	.341(**)	.01057(**)	-.146(**)	.088(**)	0.007	0.016
CRACKWATER	.029(**)	-.041 (**)	1 .184 (**)	-.035(**)	.01 .184 (**)	.026(**)		-0.01	.032(**)
CREQUESANI	.045(**)	.200(**)	-.114(**)	.041(**)				-0.018	.039(**)
AREQUENUCO	.049(**)			.116(**)		-.050 (**)			.112(**)
AREQUEMEMB	-.069 (**)			.080(**)		.456(**)			.051(**)
VREQUEREND	.056(**)			.082(**)		.089(**)			.027(**)
VREQUEPERC	.088(**)	1		.056(**)		-.188(**)			0.004
EGRAUIDESC				-.182(**)		.023(*)			-.113(**)
GRAUIDSCC	.088(**)			-.046 (**)	1	.026 (**)			-.114(**)
CREQUEITY	-.182(**)					-.311(**)			.366(**)
QREQUEQTPD	-.041(**)			0.01	1	.592(**)	1		0.002
QREQUE0A15	.023(*)			-.311(**)	.592(**)				-.178(**)
QREQUEAP16	-.071(**)			.298(**)	.673(**)	-.198(**)			.166(**)
NREQUEIDDC	-.113(**)			.366(**)	0.002	-.178(**)	.166(**)		1

Table 5.26 Result of Correlation Analysis by Pearson Coefficient.

Appendix F

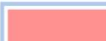
Attribute	Gini Distribution				
CREQUEORGA	0.2154	Values	Count	Percent	Histogram
		SSP	9478	88.07 %	
		SDS	987	9.17%	
		DPT	1	0.01 %	
		SP	3	0.03 %	
		ITB	7	0.07%	
		SSD	1	0.01 %	
		SPP	1	0.01 %	
		RFB	3	0.03 %	
		ITP	1	0.01 %	
		SSO	2	0.02 %	
		RGN	1	0.01 %	
		MEX	1	0.01 %	
		DS	1	0.01 %	

		N/A	268	2.49 %	
		AP	1	0.01 %	
		DPD	1	0.01 %	
		SSS	1	0.01 %	
		OE	1	0.01 %	
		SDSP	1	0.01 %	
		SSPQ	1	0.01 %	
		SEPC	1	0.01 %	

CTPLGDCODI 0.3060		Values	Count	Percent	Histogram
		R	8910	82.79 %	
		EST	50	0.46 %	
		1ATV	28	0.26 %	
		FAV	1	0.01 %	
		AV	709	6.59 %	
		TV	689	6.40 %	
		SUB	10	0.09 %	
		CRG	60	0.56 %	
		ROD	16	0.15 %	

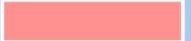
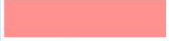
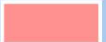
		VL	39	0.36 %	
		AT	66	0.61 %	
		LD	30	0.28 %	
		BC	59	0.55 %	
		ESC	35	0.33 %	
		LRG	3	0.03 %	
		2ATV	21	0.20 %	
		3ATV	11	0.10 %	
		PRL	1	0.01 %	
		PRAÇA	7	0.07%	
		6ATV	1	0.01 %	
		2ASUB	2	0.02 %	
		5ATV	1	0.01 %	
		LOT	1	0.01 %	
		SIT	3	0.03 %	
		1ASUB	1	0.01 %	
		Q	1	0.01 %	
		4ATV	3	0.03 %	

		w	1	0.01 %	
		MRO	1	0.01 %	
		3ASUB	1	0.01 %	
		PTE	1	0.01 %	

NESTANNOMC	0.3918	Values	Count	Percent	Histogram
		N/A	7908	73.48 %	
		PERNAMBUCO	2812	26.13 %	
		PARAIBA	13	0.12 %	
		ALAGOAS	9	0.08 %	
		RIOGRANDEDONORTE	2	0.02 %	
		SÃO PAULO	6	0.06 %	
		TO	1	0.01 %	
		BAHIA	3	0.03 %	
		RIO DE JANEIRO	3	0.03 %	
		PIAUI	1	0.01 %	
		MINAS GERAIS	1	0.01 %	
		CEARA	2	0.02 %	
		SERGIPE	1	0.01 %	

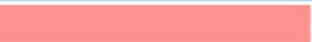
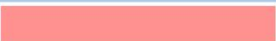
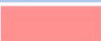
ESITESDESC 0.0424		Values	Count	Percent	Histogram
		N/A	10530	97.84 %	
		CARRIER OF CHRONIC DISEASE	104	0.97%	
		DISABLED PERSON UNABLE TO PROVIDE FOR HIMSELF 121		1.12 %	
		CHILD/TEENAGER UNDER SOCIO-EDUCATIONAL MEASURES	5	0.05 %	
		CHILD/ADOLESCENT UNDER SPECIAL PROTECTION	1	0.01 %	
		CHILDREN AGED 0 TO 6 WITH MALNUTRITION	1	0.01 %	

ETPPISDESC 0.6506		Values	Count	Percent	Histogram
		SUBFLOOR	3466	32.21 %	
		BRICK/CEMENT 5072	47.13 %		
		TERRABATIDA	1498	13.92 %	
		CERAMICS	700	6.50 %	
		STILTS	26	0.24 %	

ETPCOBDESC	0.6818				
ETPCOBDESC	0.6818	Values	Count	Percent	Histogram
		SLAB	2183	20.28 %	
		CERAMICS TILES	5085	47.25 %	
		ZINC/BRAZILIT	2070	19.23 %	
		PLASTIC/CARNABY	1396	12.97 %	
		IT DOES NOT HAVE	28	0.26 %	
EGLETEDESC	0.7074	Values	Count	Percent	Histogram
		TENANT	1567	14.56 %	
		OWNER	4725	43.90 %	
		PROPPARENT	1831	17.01 %	
		TENANT	2385	22.16 %	
		PROPPATRON	254	2.36 %	
ETPCONDESC	0.7614	Values	Count	Percent	Histogram
		FINISHING	2795	25.97 %	
		READY	3264	30.33 %	

		PARALYZED	2815	26.16 %	
		SEMINFORMATION	827	7.68%	
		STARTED	512	4.76%	
		OTHERS	549	5.10 %	

Appendix G

Attribute	Gini	Distribution			
CREQUEORGA	0.2101	Values	Count	Percent	Histogram
		SSP	9478	88.07 %	
		OTHERS	1284	11.93 %	
CTPLGDCODI	0.3043	Values	Count	Percent	Histogram
		R	8910	82.79 %	
		OTHERS	454	4.22 %	
		AV	709	6.59 %	
		TV	689	6.40 %	
NESTANNOMC	0.3918	Values	Count	Percent	Histogram
		N/A	7908	73.48 %	
		PERNAMBUCO	2812	26.13 %	
		OTHERS	42	0.39 %	
ESITESDESC	0.0422	Values	Count	Percent	Histogram
		N/A	10530	97.84 %	
		OTHERS	232	2.16 %	